

ML Hybrid Models Comparison Report

Generated: 2025-11-23T22:31:37.993434

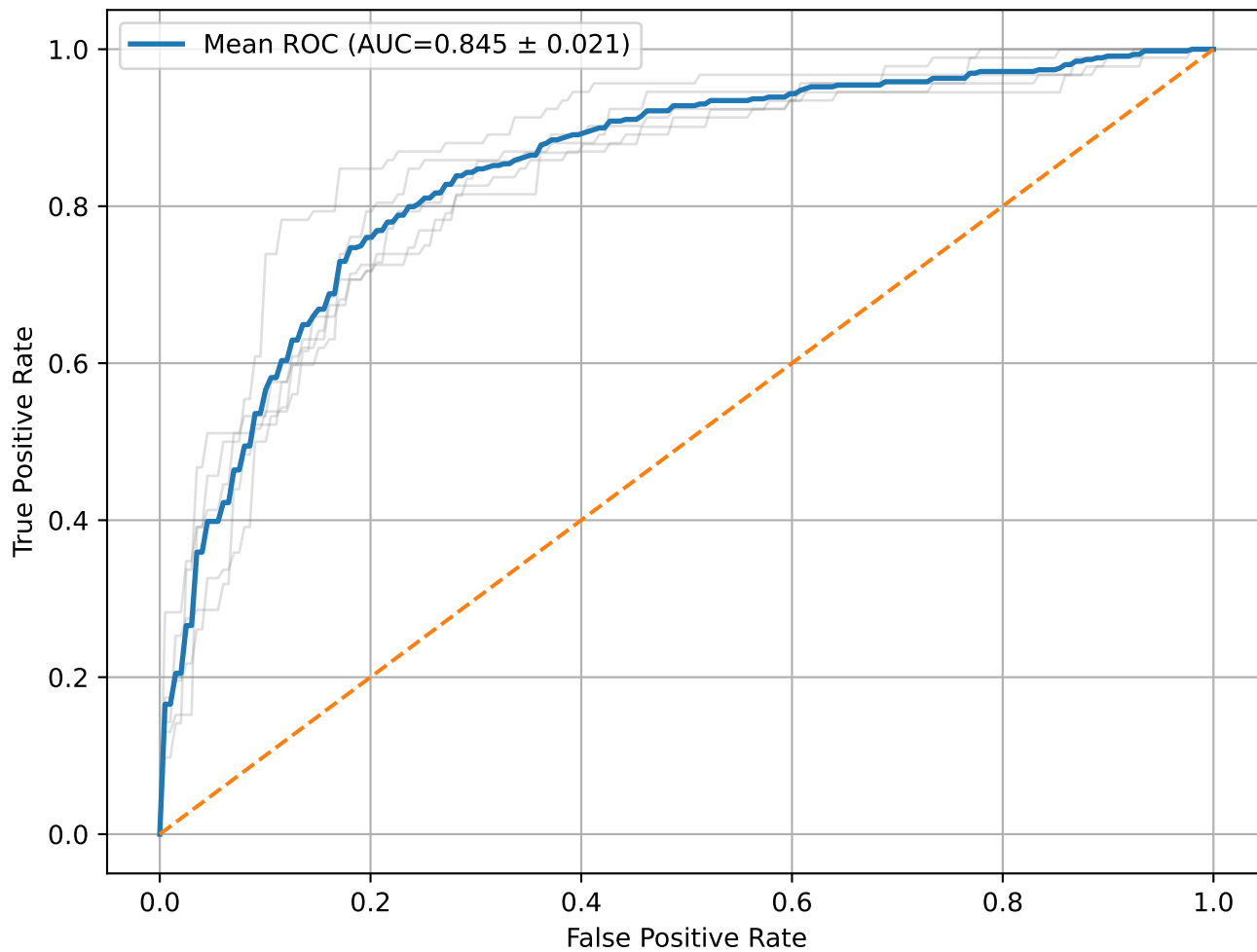
Dataset samples: 906, features: 247

Models: logistic, rf, svm, xgb, lgb, dnn, stacking, voting, bagging, pca_svm, ga_xgb, pso_svm, bayes_opt_svms

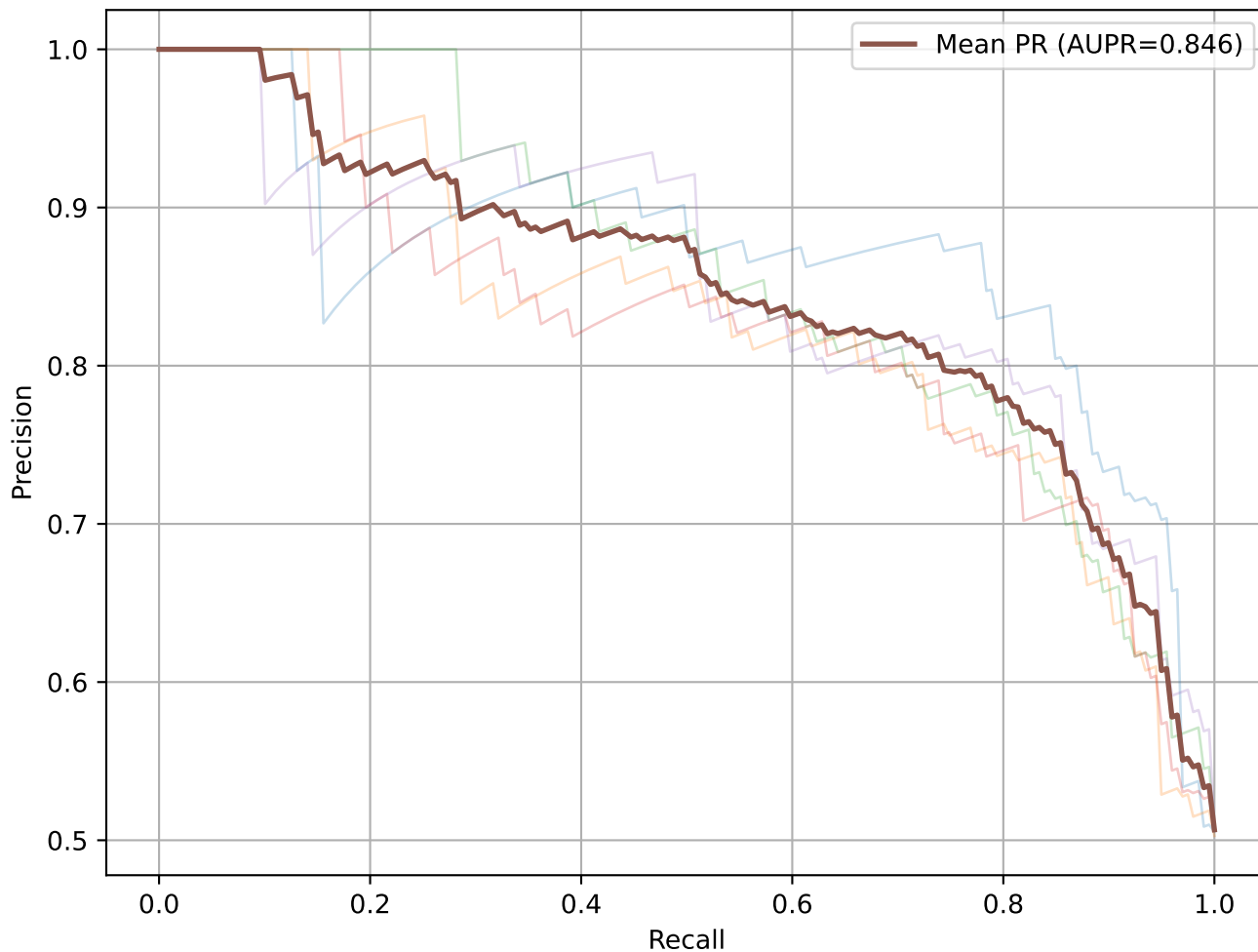
StratifiedKFold: 5 splits

Random seed: 42

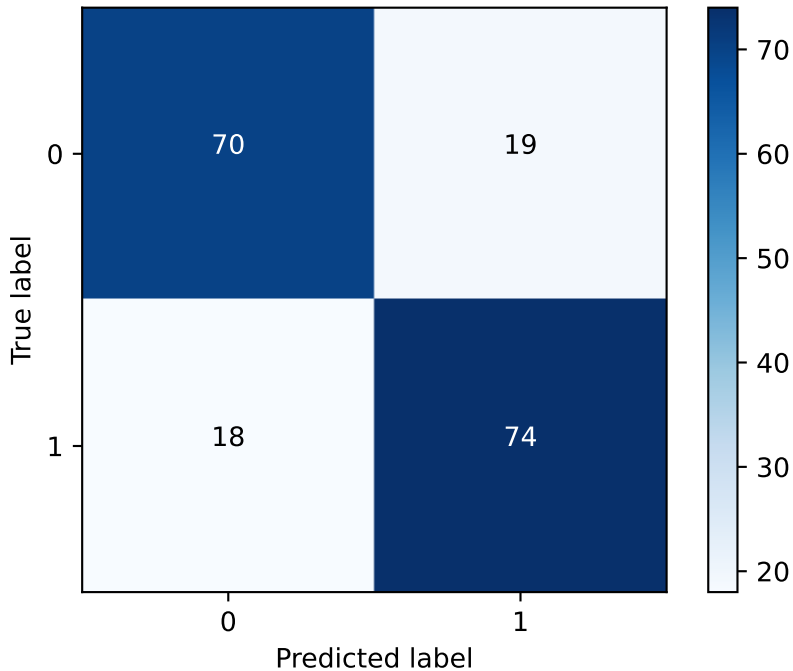
ROC - LOGISTIC



Precision-Recall - LOGISTIC



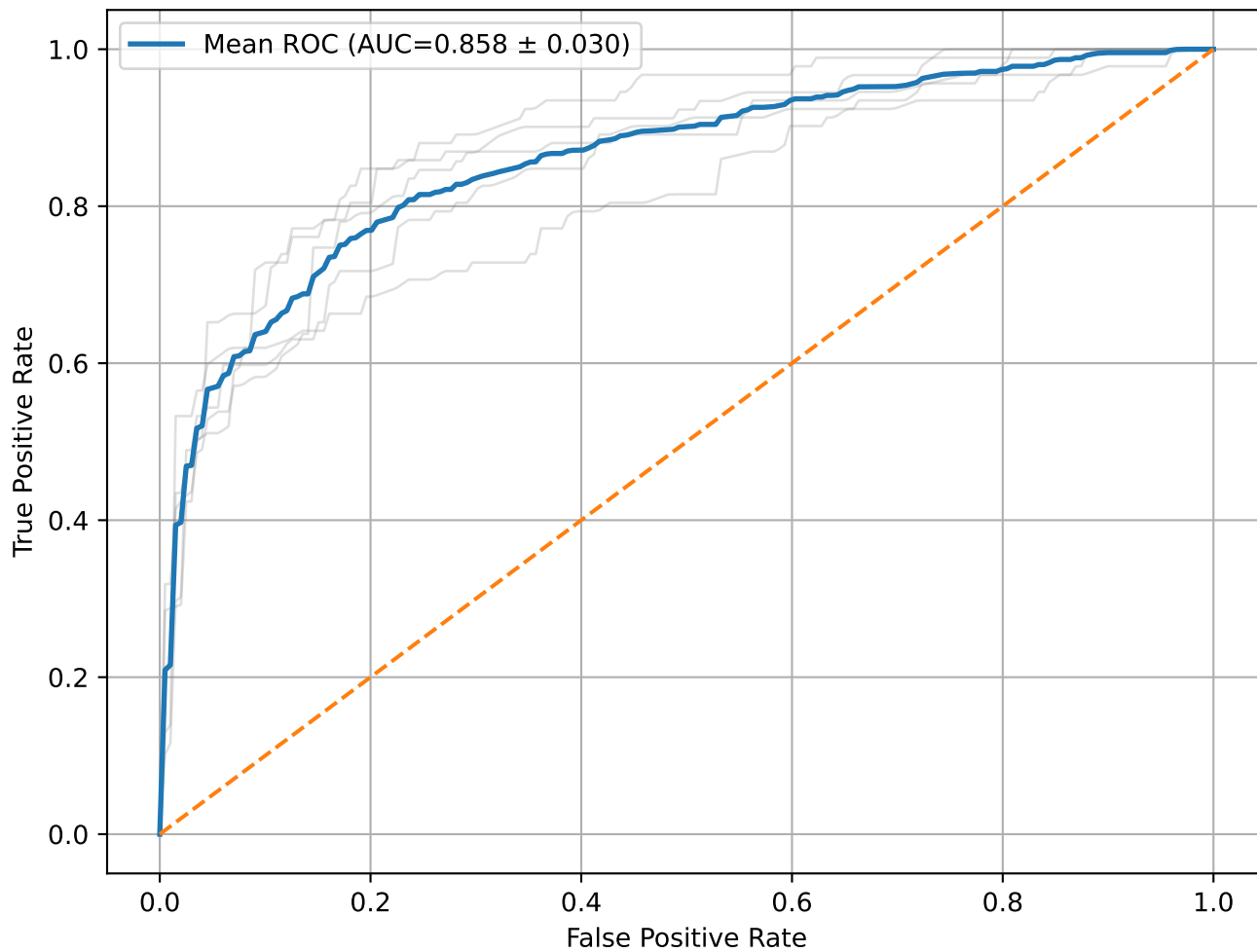
Confusion Matrix - LOGISTIC (last fold)



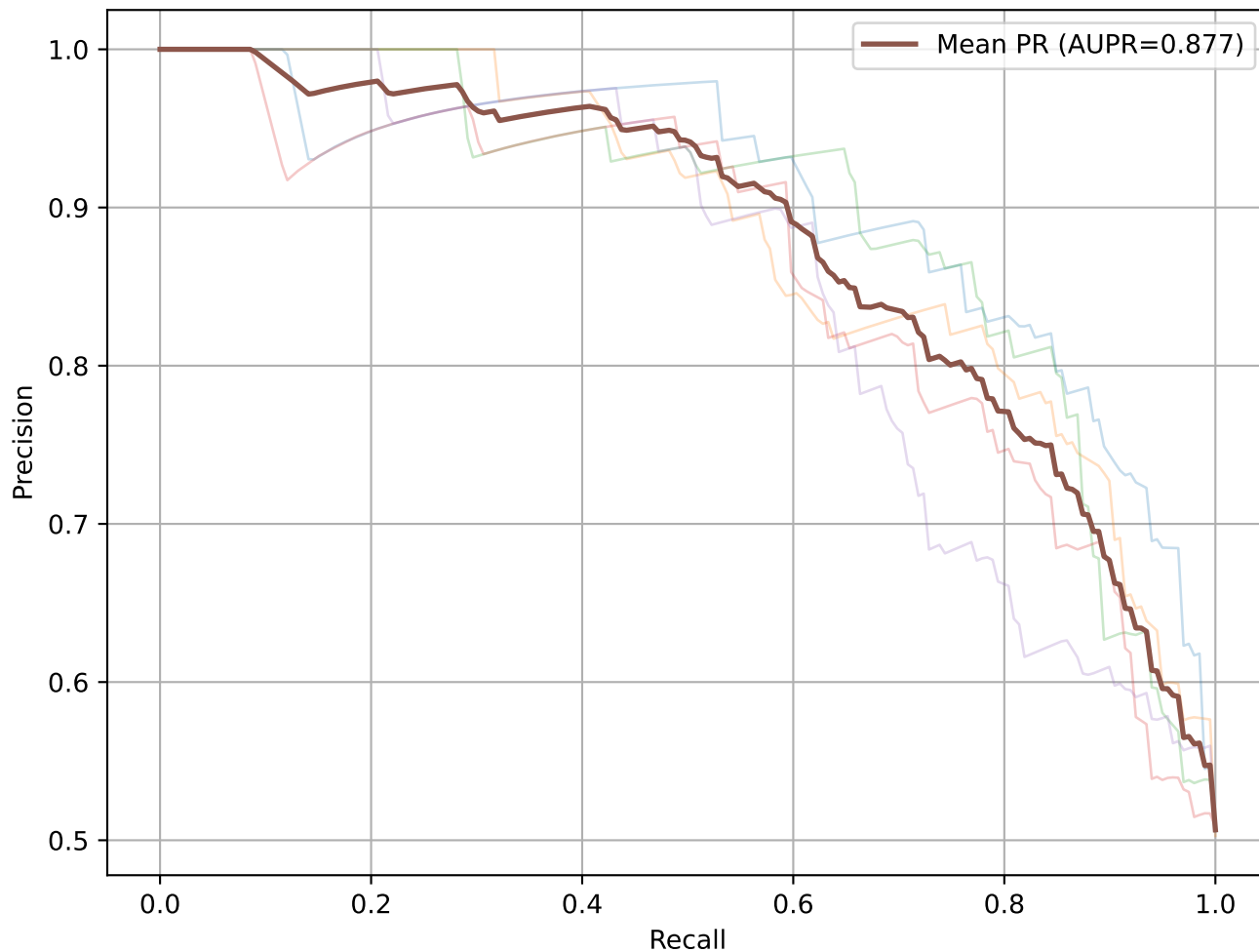
ClassificationReport - LOGISTIC (last fold)

	precision	recall	f1-score	support
0	0.80	0.79	0.79	89
1	0.80	0.80	0.80	92
accuracy			0.80	181
macro avg	0.80	0.80	0.80	181
weighted avg	0.80	0.80	0.80	181

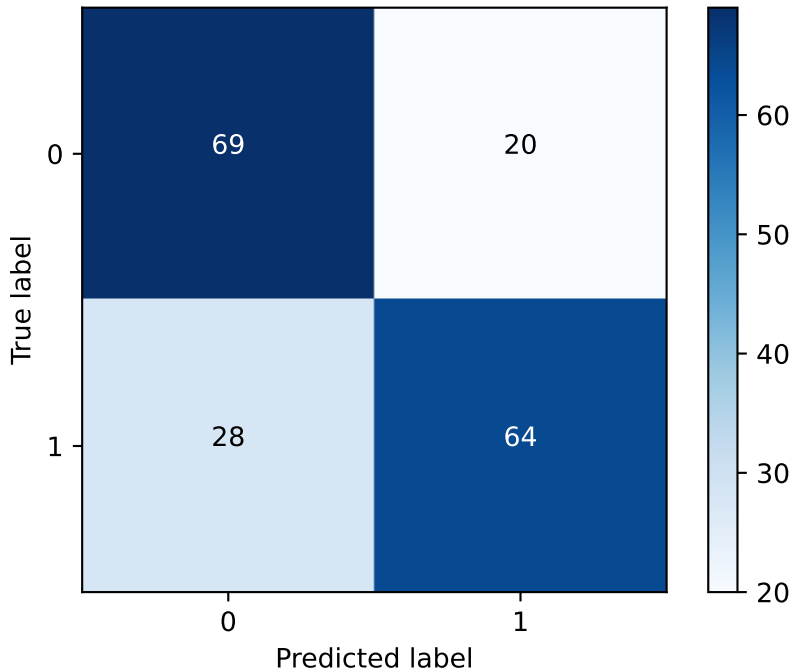
ROC - RF



Precision-Recall - RF



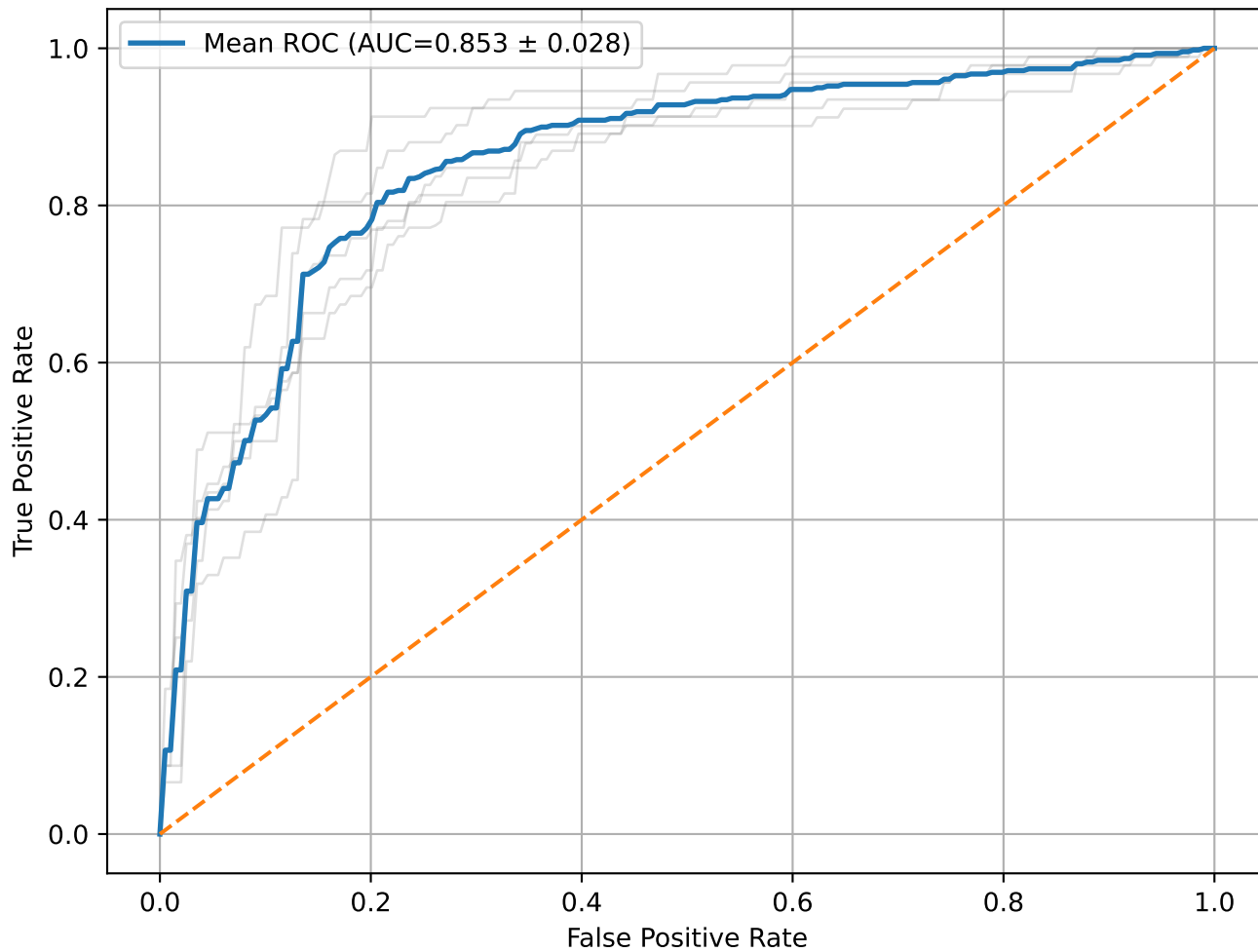
Confusion Matrix - RF (last fold)



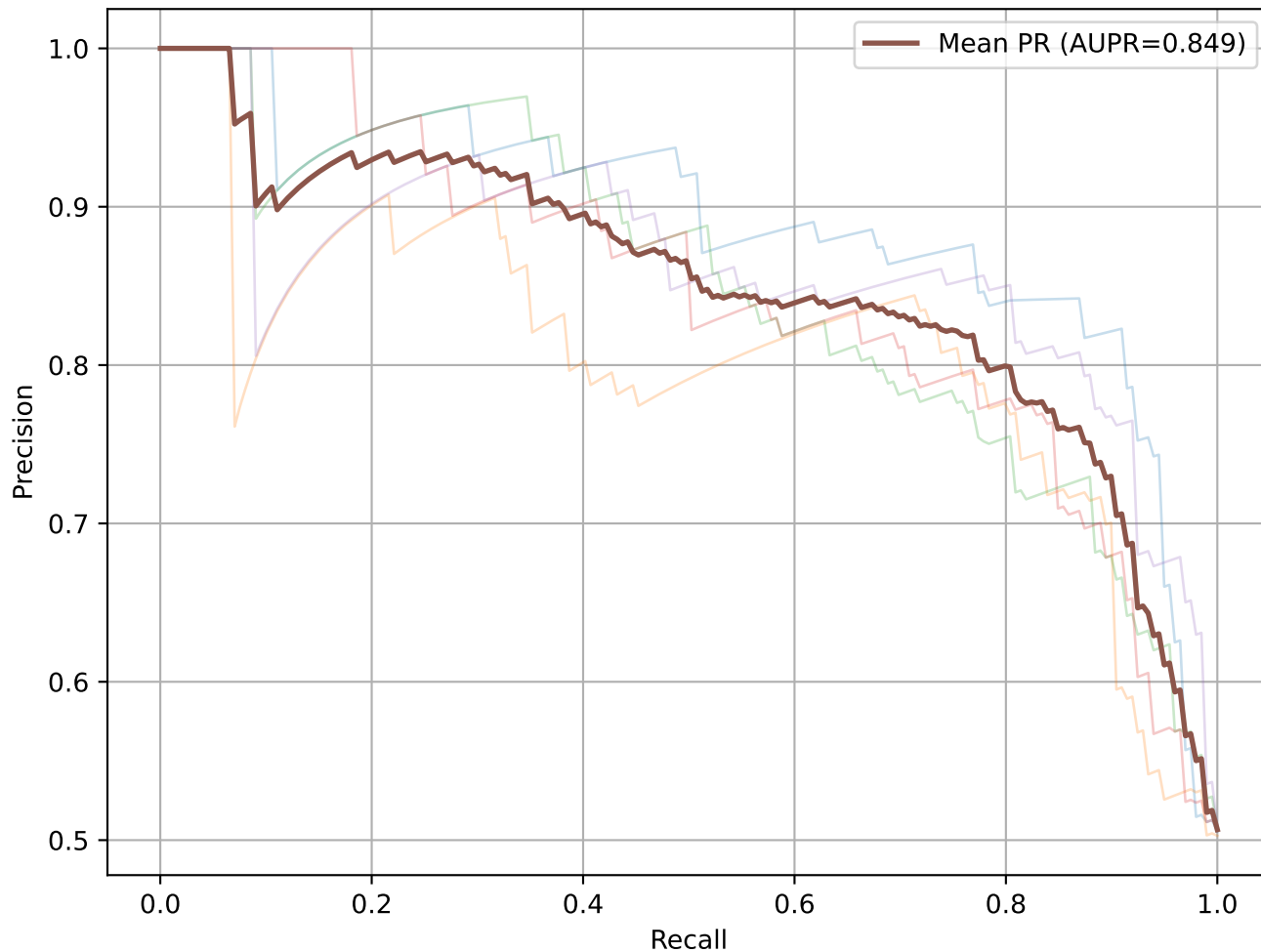
ClassificationReport - RF (last fold)

	precision	recall	f1-score	support
0	0.71	0.78	0.74	89
1	0.76	0.70	0.73	92
accuracy			0.73	181
macro avg	0.74	0.74	0.73	181
weighted avg	0.74	0.73	0.73	181

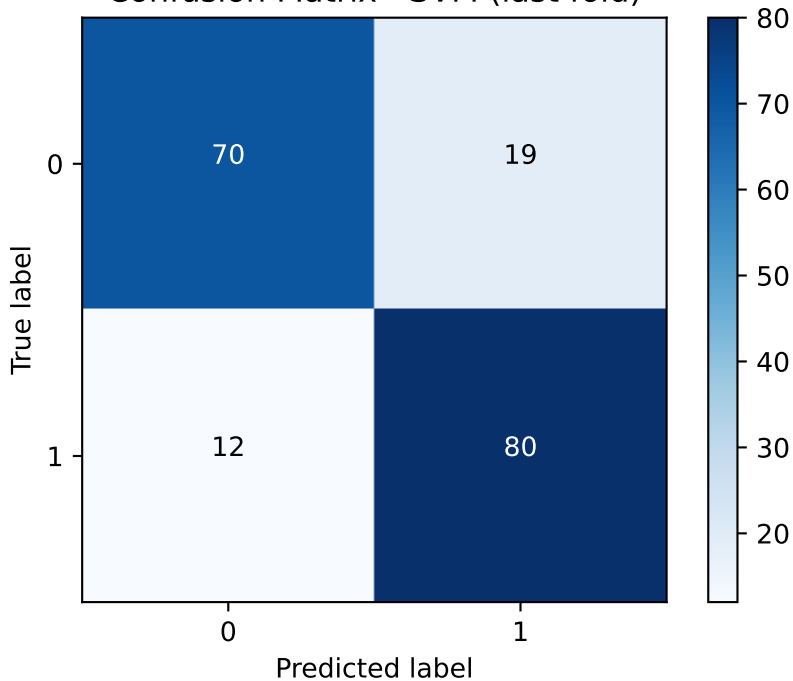
ROC - SVM



Precision-Recall - SVM



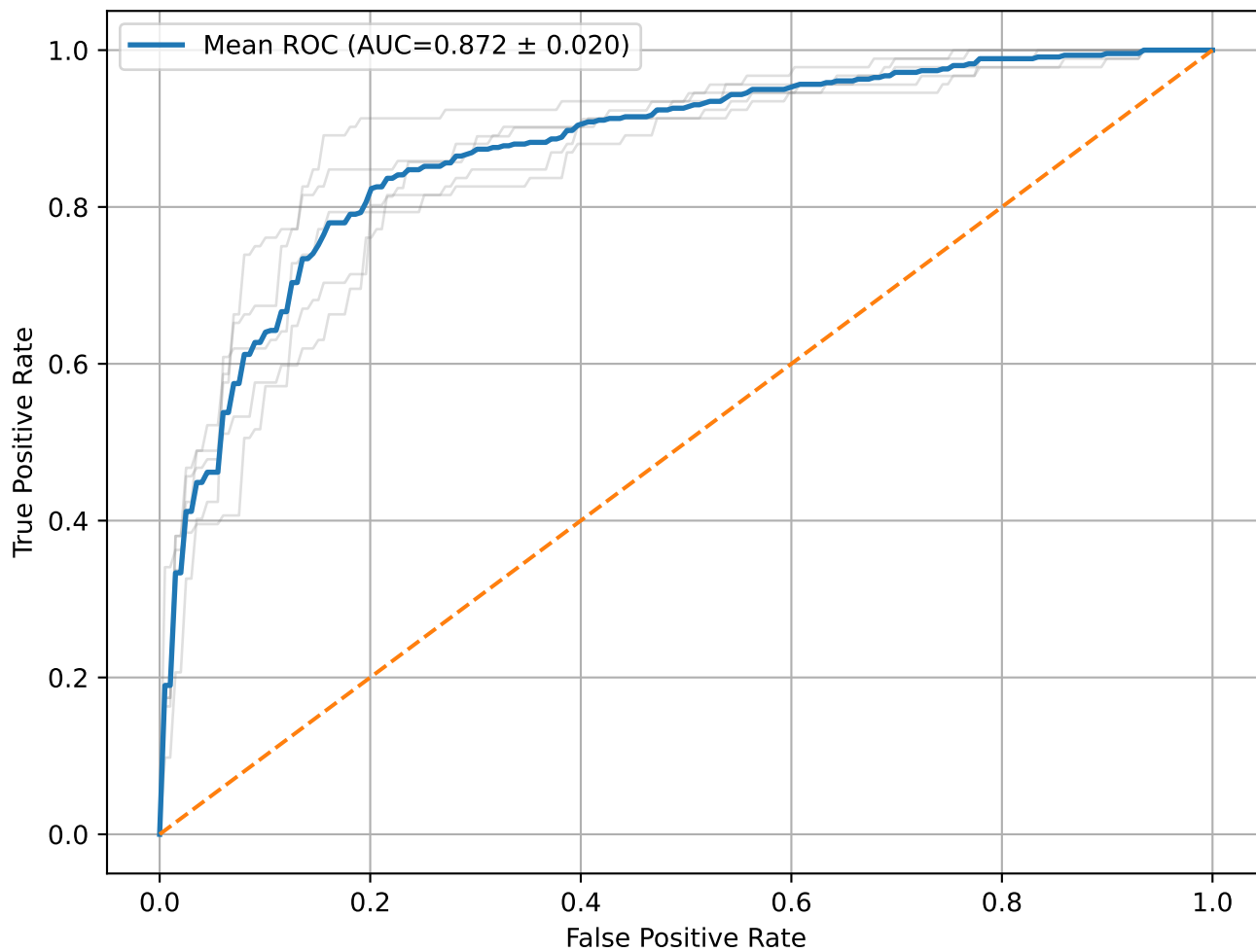
Confusion Matrix - SVM (last fold)



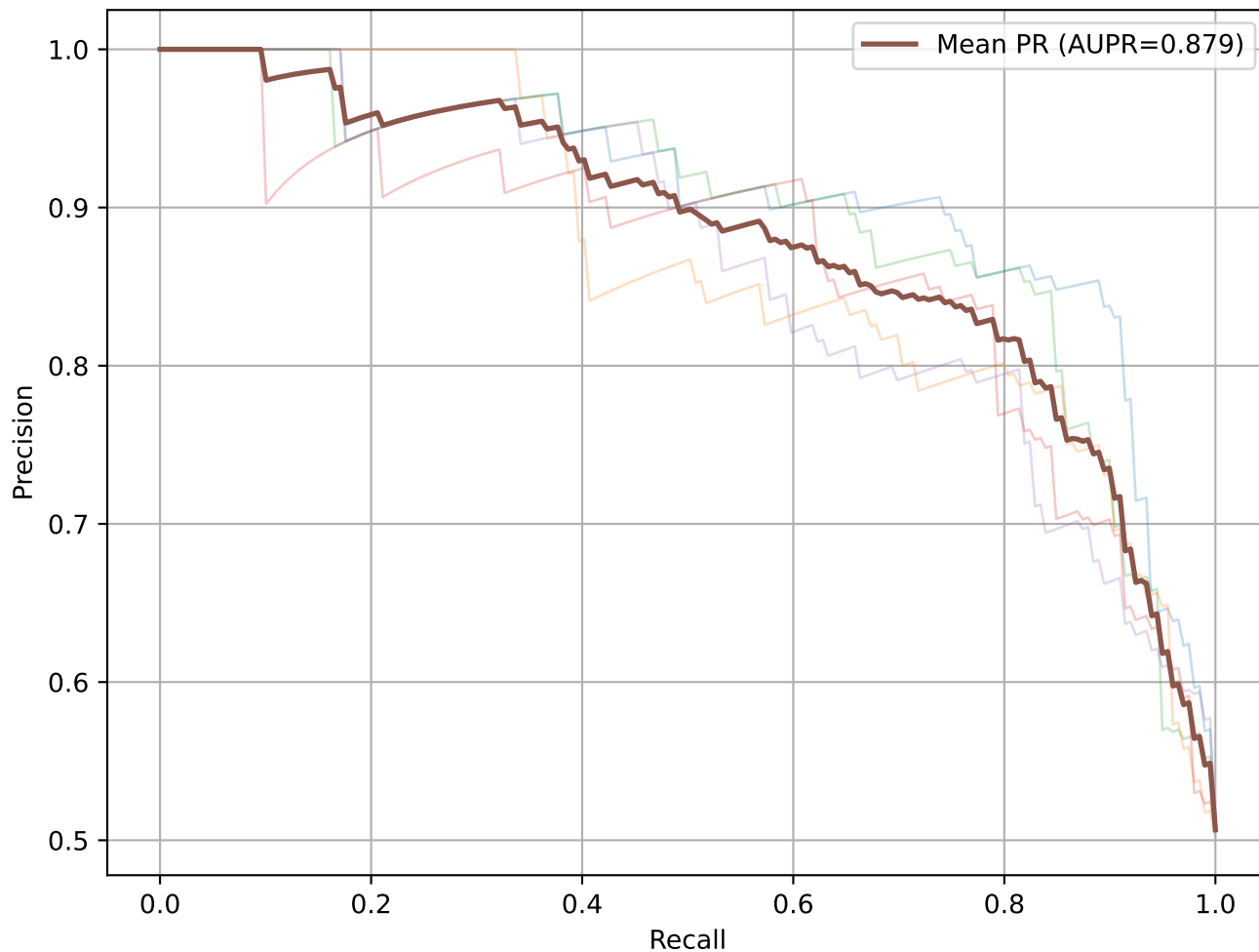
ClassificationReport - SVM (last fold)

	precision	recall	f1-score	support
0	0.85	0.79	0.82	89
1	0.81	0.87	0.84	92
accuracy			0.83	181
macro avg	0.83	0.83	0.83	181
weighted avg	0.83	0.83	0.83	181

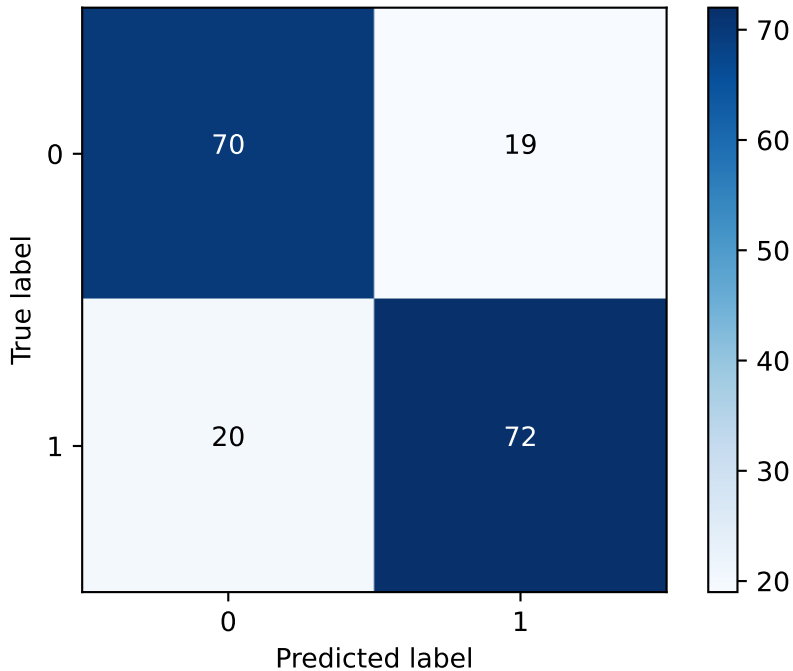
ROC - XGB



Precision-Recall - XGB



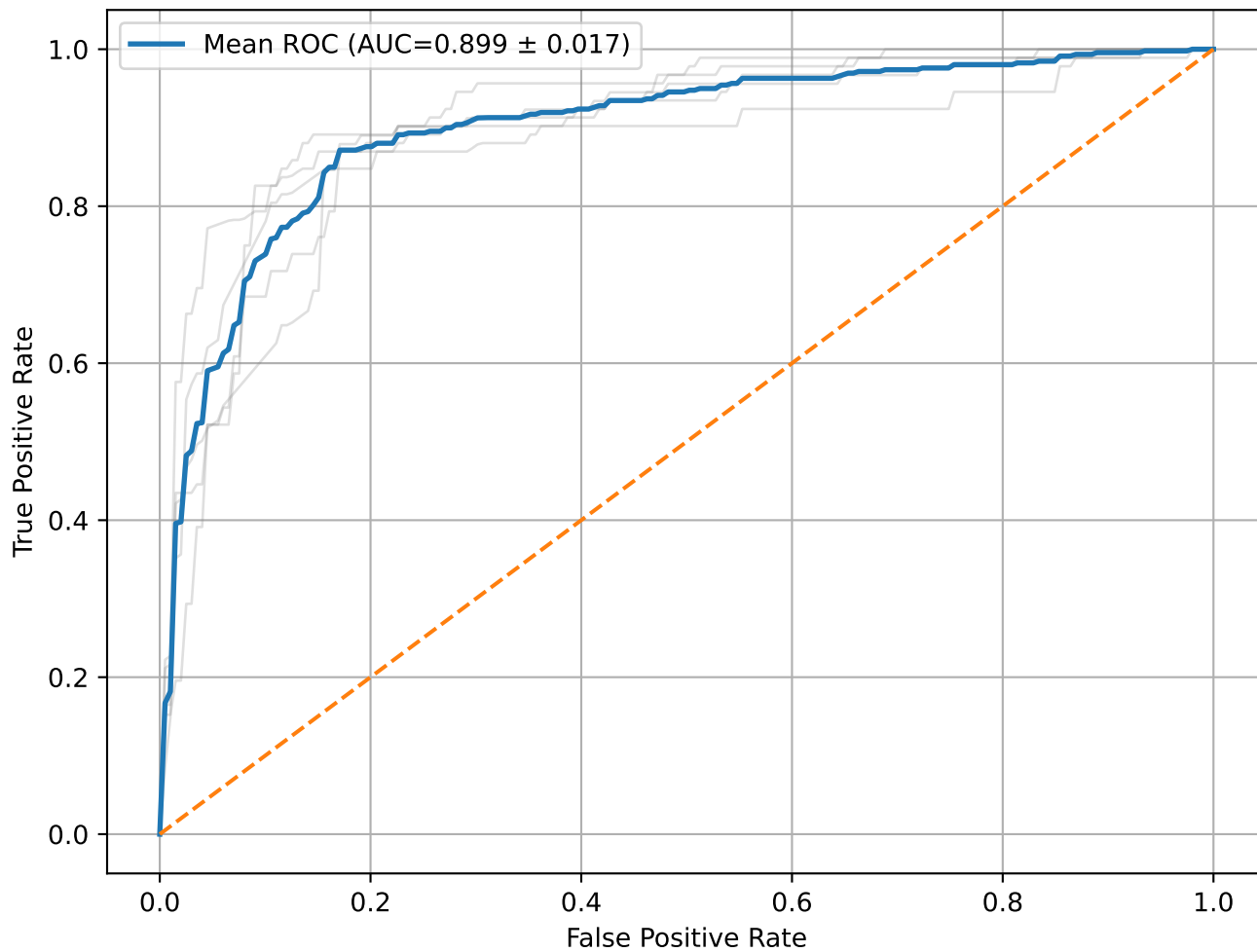
Confusion Matrix - XGB (last fold)



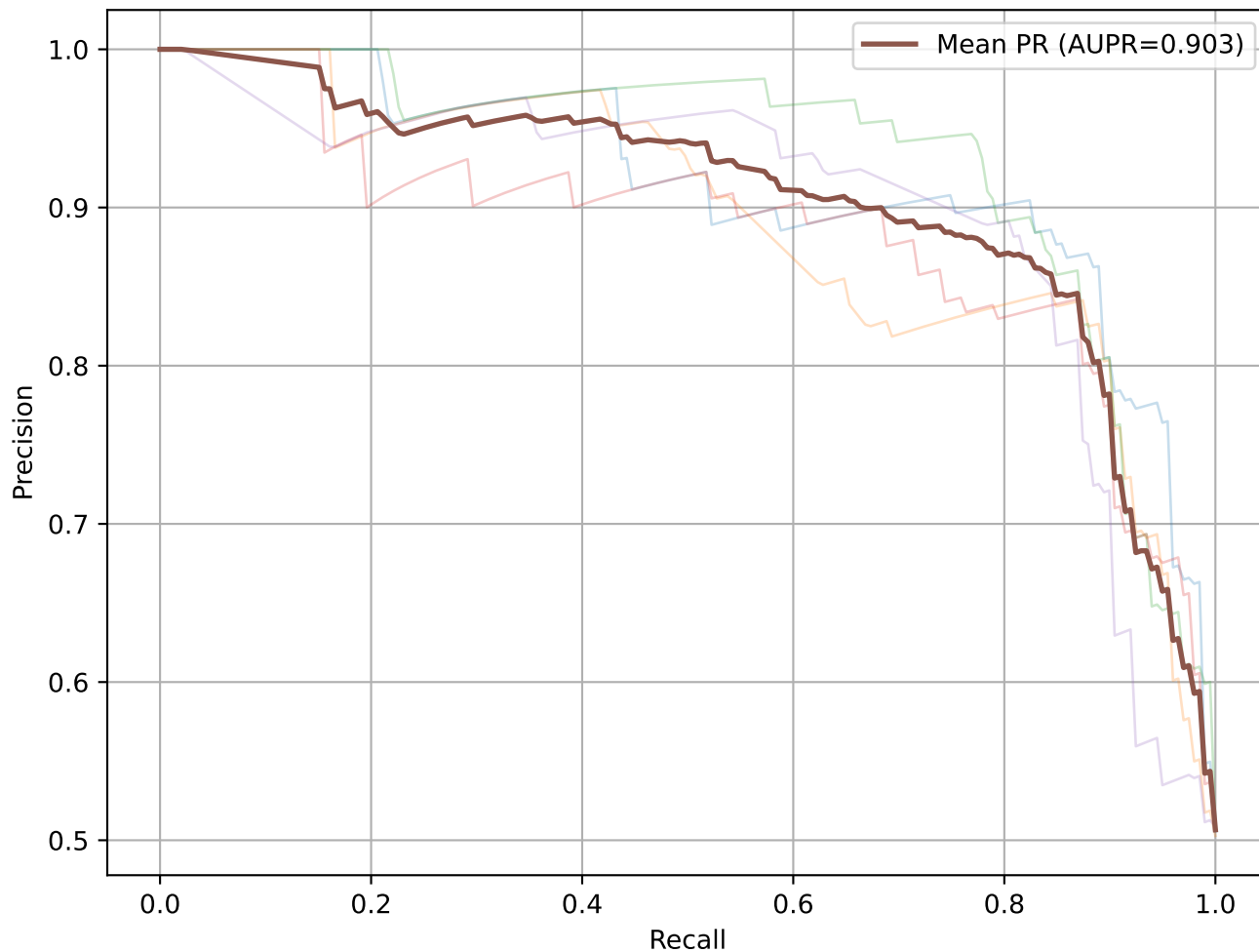
ClassificationReport - XGB (last fold)

	precision	recall	f1-score	support
0	0.78	0.79	0.78	89
1	0.79	0.78	0.79	92
accuracy			0.78	181
macro avg	0.78	0.78	0.78	181
weighted avg	0.78	0.78	0.78	181

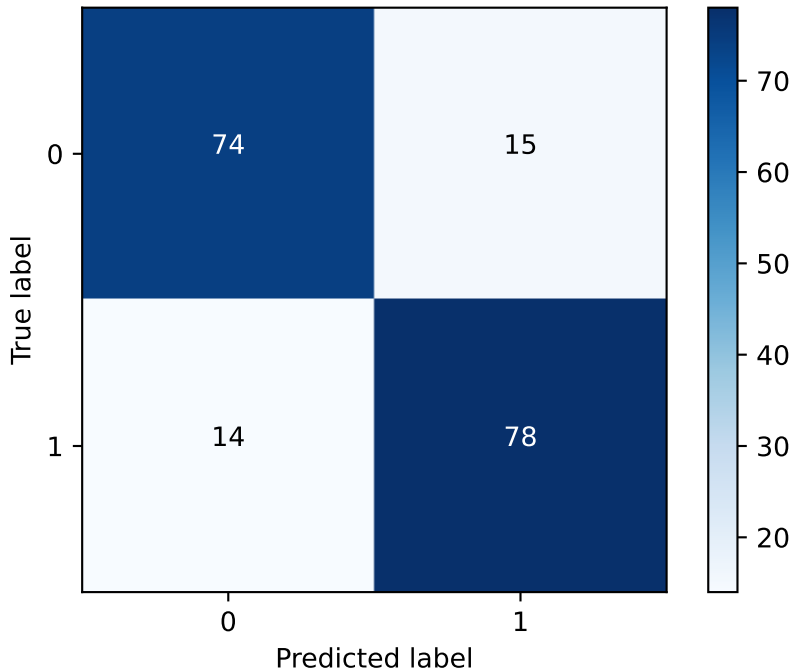
ROC - LGB



Precision-Recall - LGB



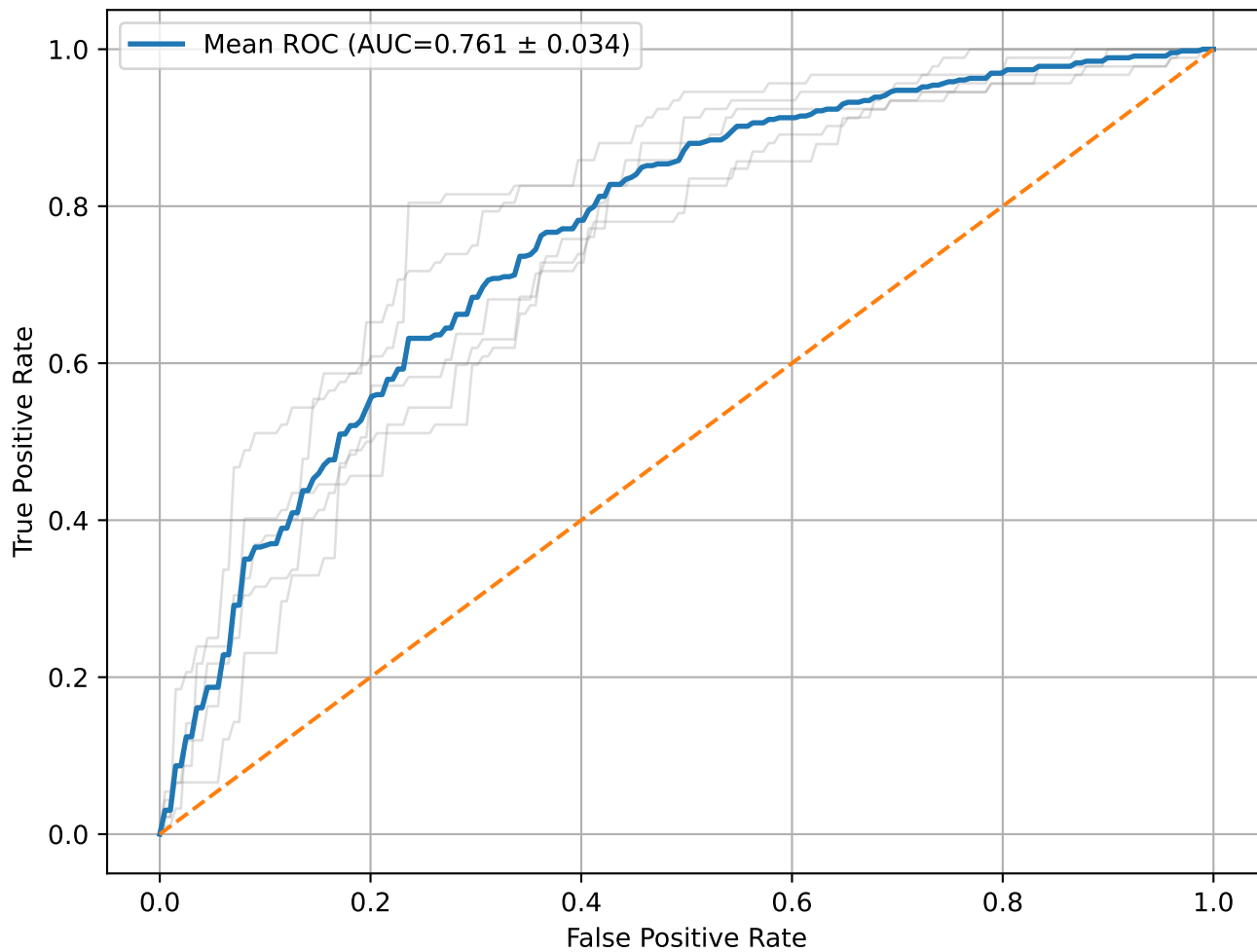
Confusion Matrix - LGB (last fold)



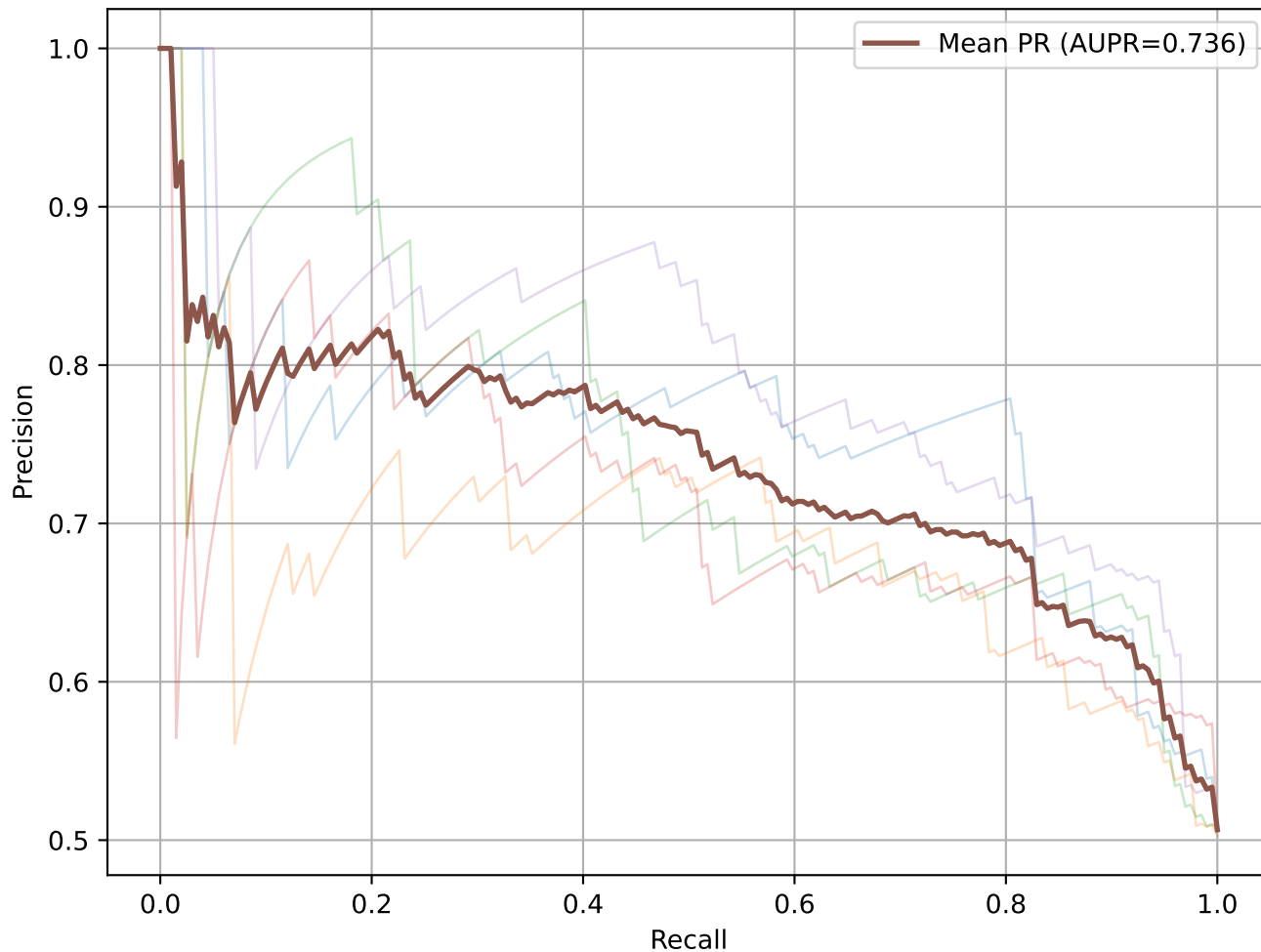
ClassificationReport - LGB (last fold)

	precision	recall	f1-score	support
0	0.84	0.83	0.84	89
1	0.84	0.85	0.84	92
accuracy			0.84	181
macro avg	0.84	0.84	0.84	181
weighted avg	0.84	0.84	0.84	181

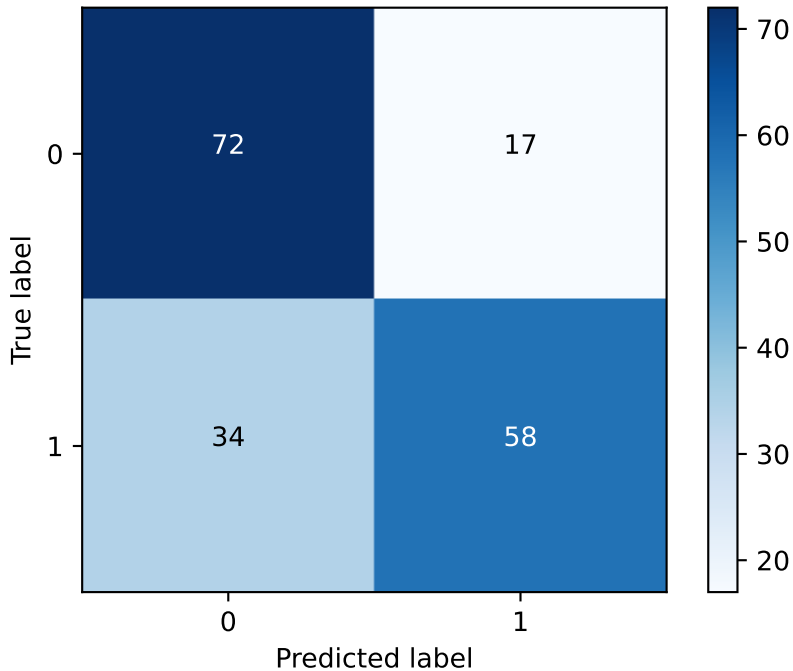
ROC - DNN



Precision-Recall - DNN



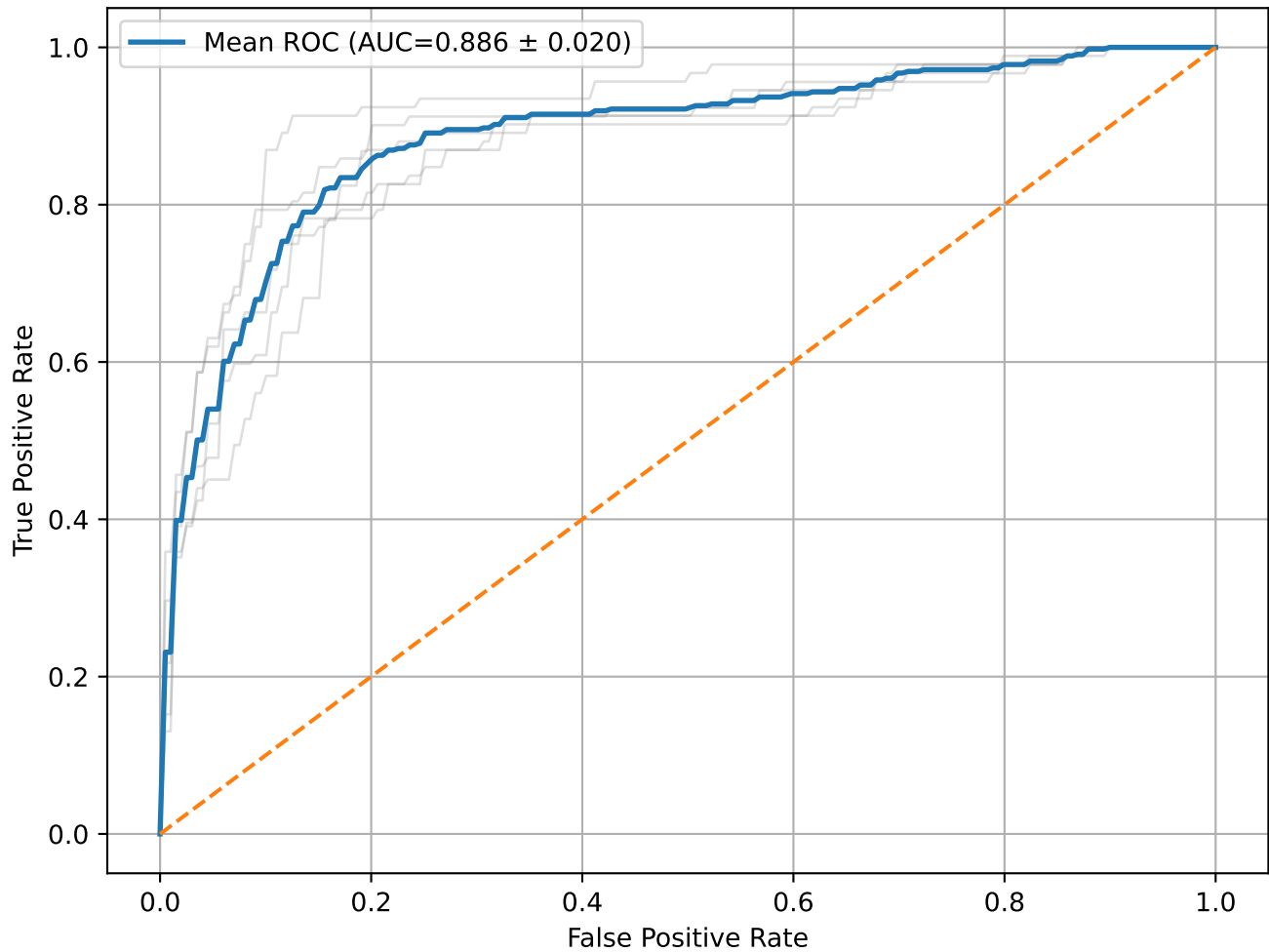
Confusion Matrix - DNN (last fold)



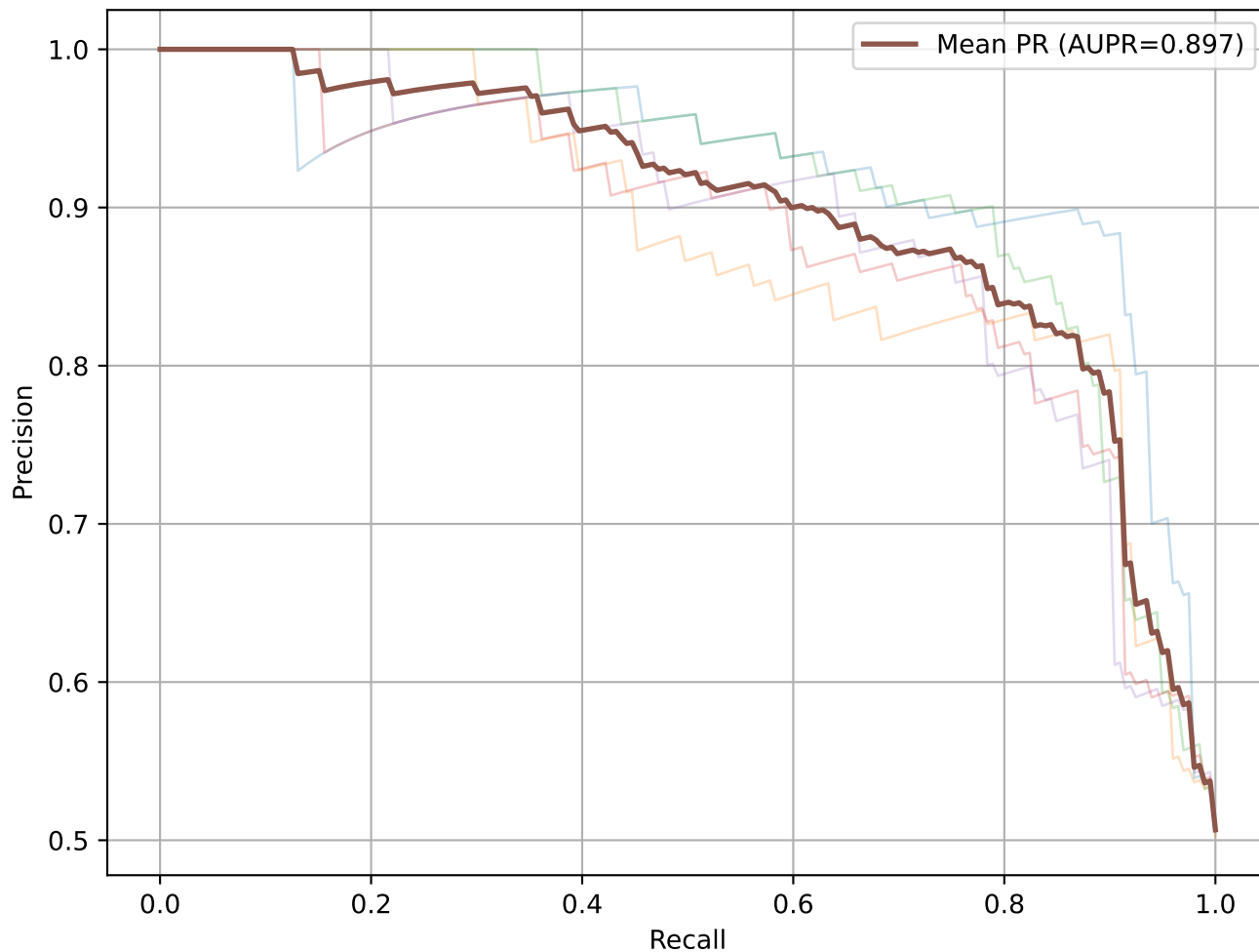
ClassificationReport - DNN (last fold)

	precision	recall	f1-score	support
0	0.68	0.81	0.74	89
1	0.77	0.63	0.69	92
accuracy			0.72	181
macro avg	0.73	0.72	0.72	181
weighted avg	0.73	0.72	0.72	181

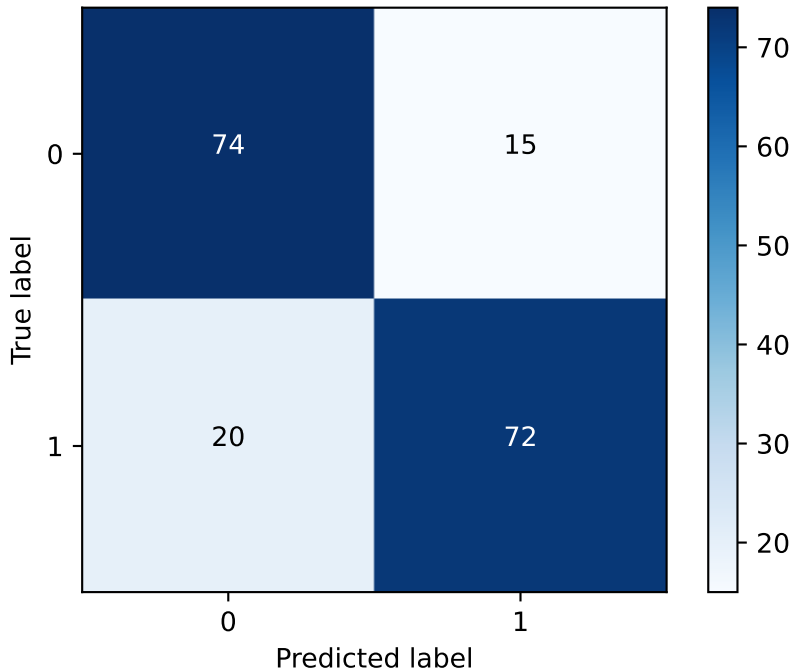
ROC - STACKING



Precision-Recall - STACKING



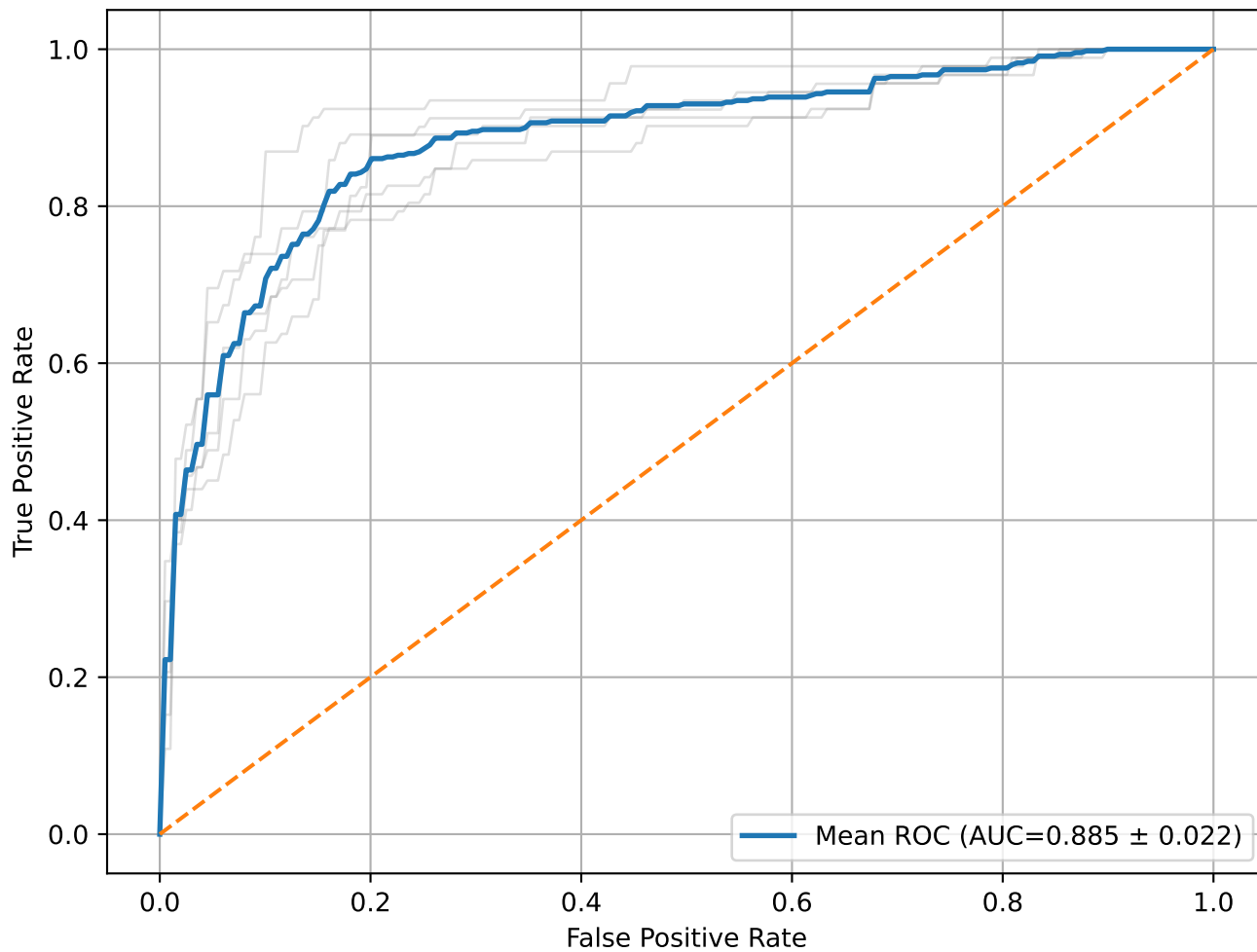
Confusion Matrix - STACKING (last fold)

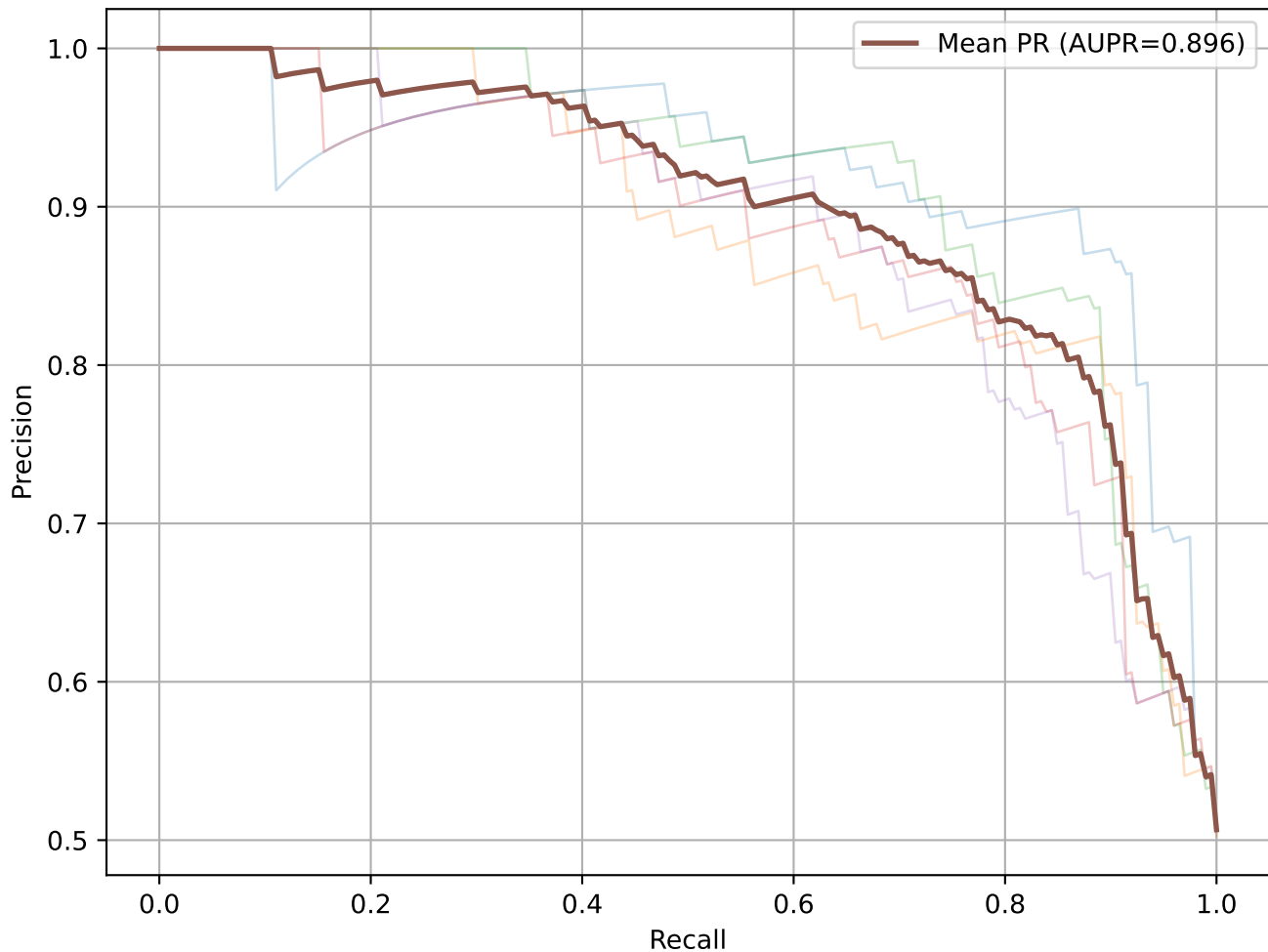


ClassificationReport - STACKING (last fold)

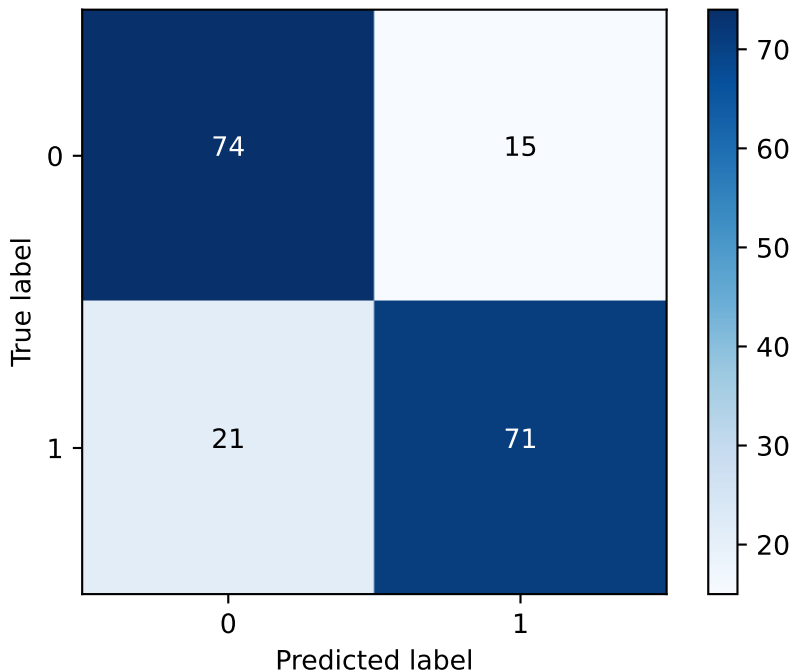
	precision	recall	f1-score	support
0	0.79	0.83	0.81	89
1	0.83	0.78	0.80	92
accuracy			0.81	181
macro avg	0.81	0.81	0.81	181
weighted avg	0.81	0.81	0.81	181

ROC - VOTING





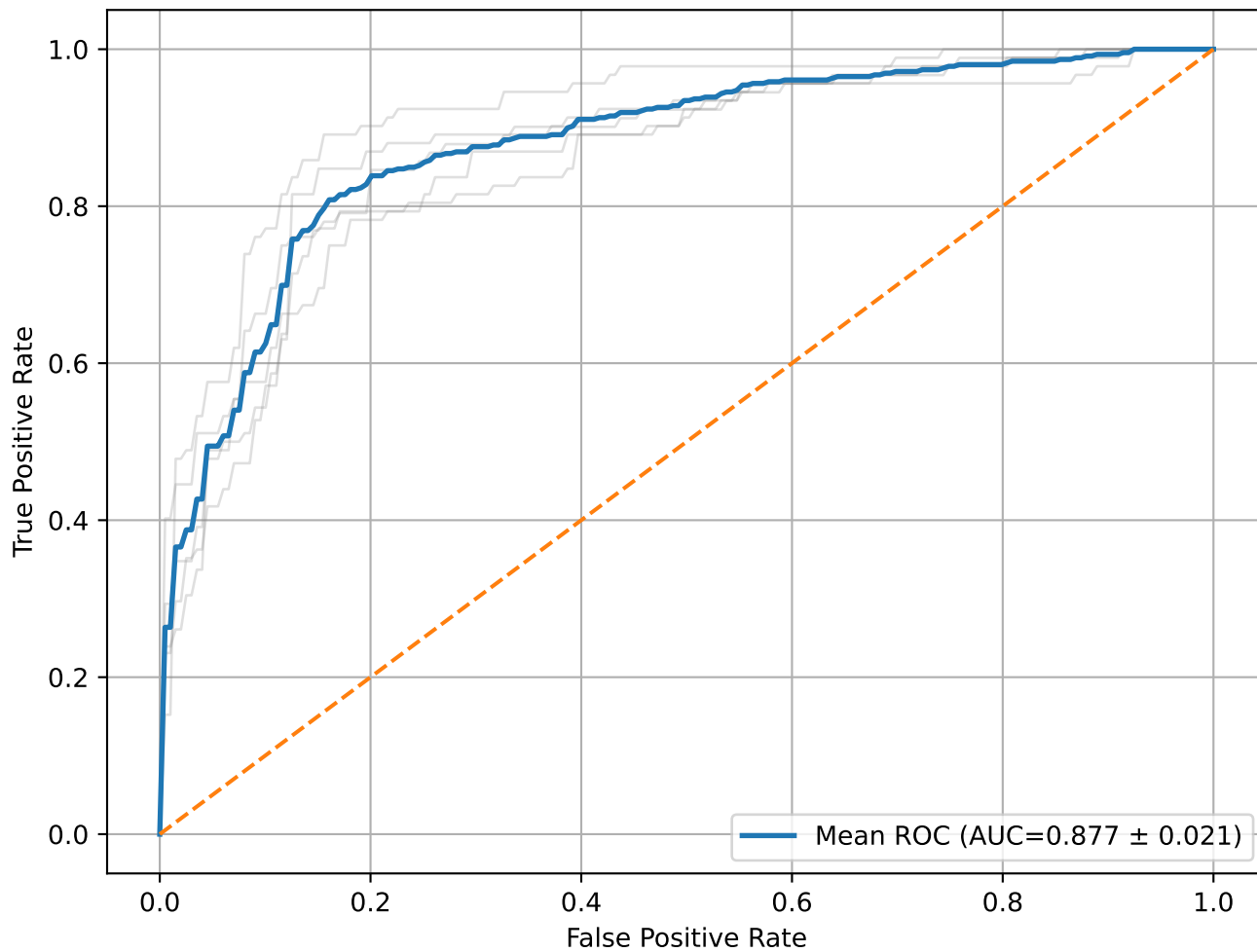
Confusion Matrix - VOTING (last fold)



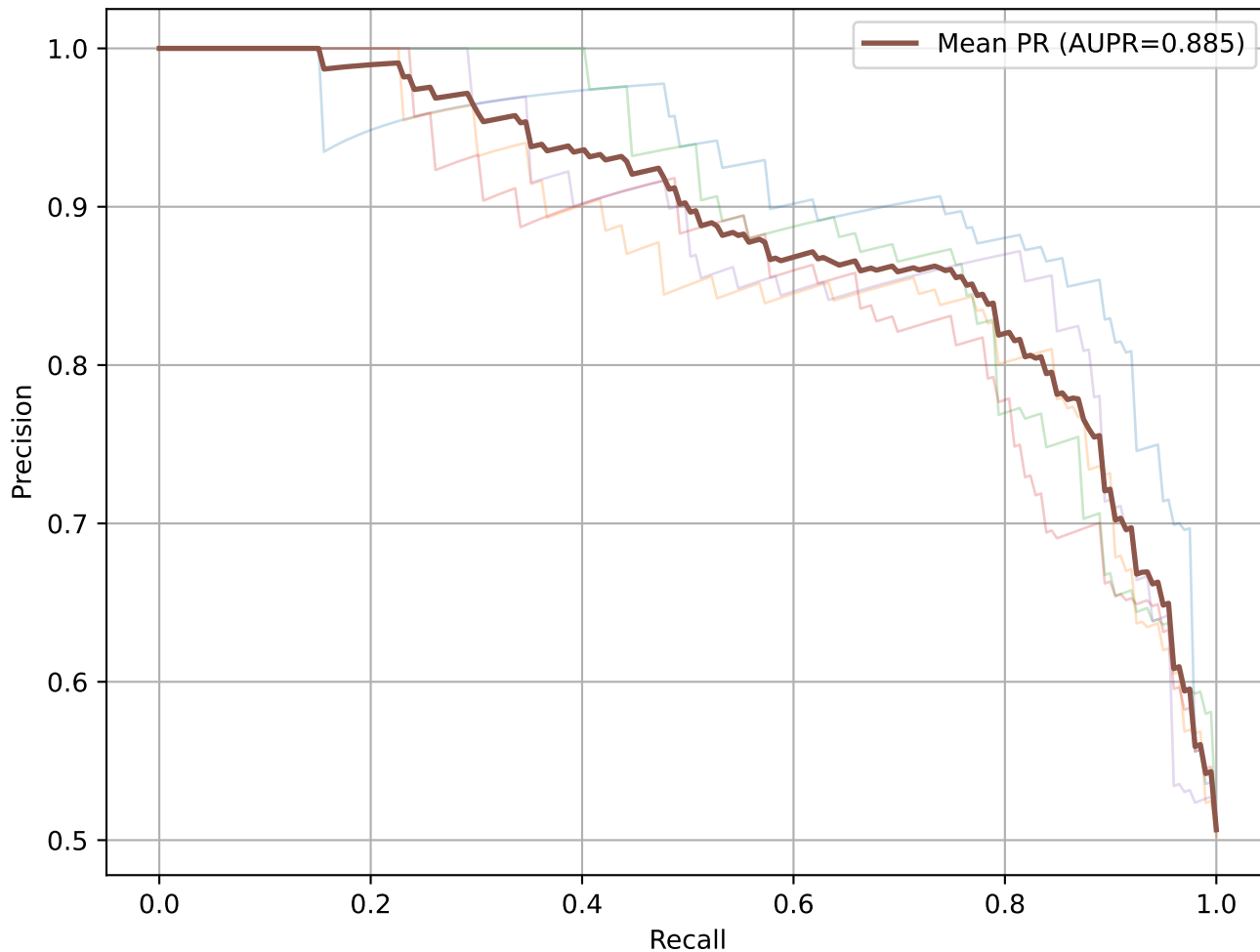
ClassificationReport - VOTING (last fold)

	precision	recall	f1-score	support
0	0.78	0.83	0.80	89
1	0.83	0.77	0.80	92
accuracy			0.80	181
macro avg	0.80	0.80	0.80	181
weighted avg	0.80	0.80	0.80	181

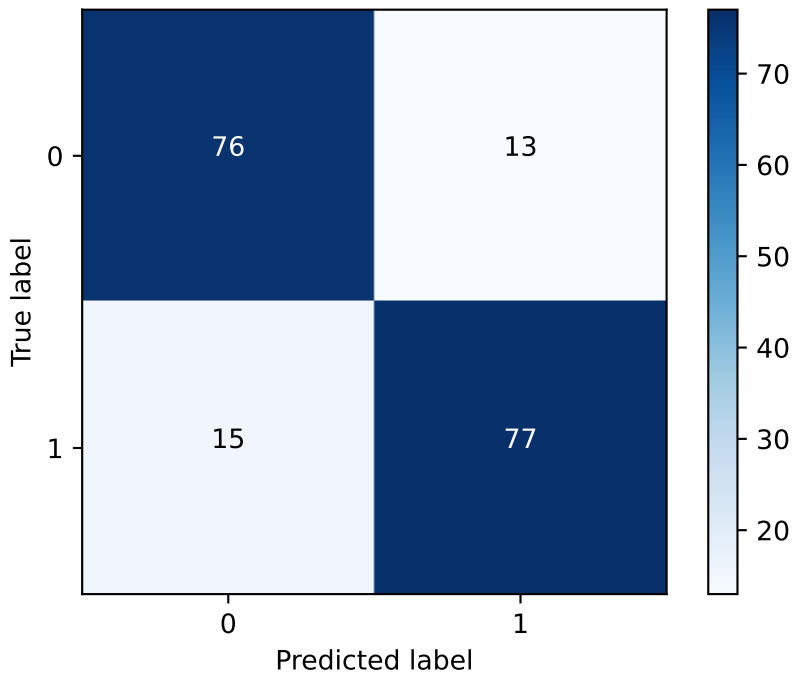
ROC - BAGGING



Precision-Recall - BAGGING



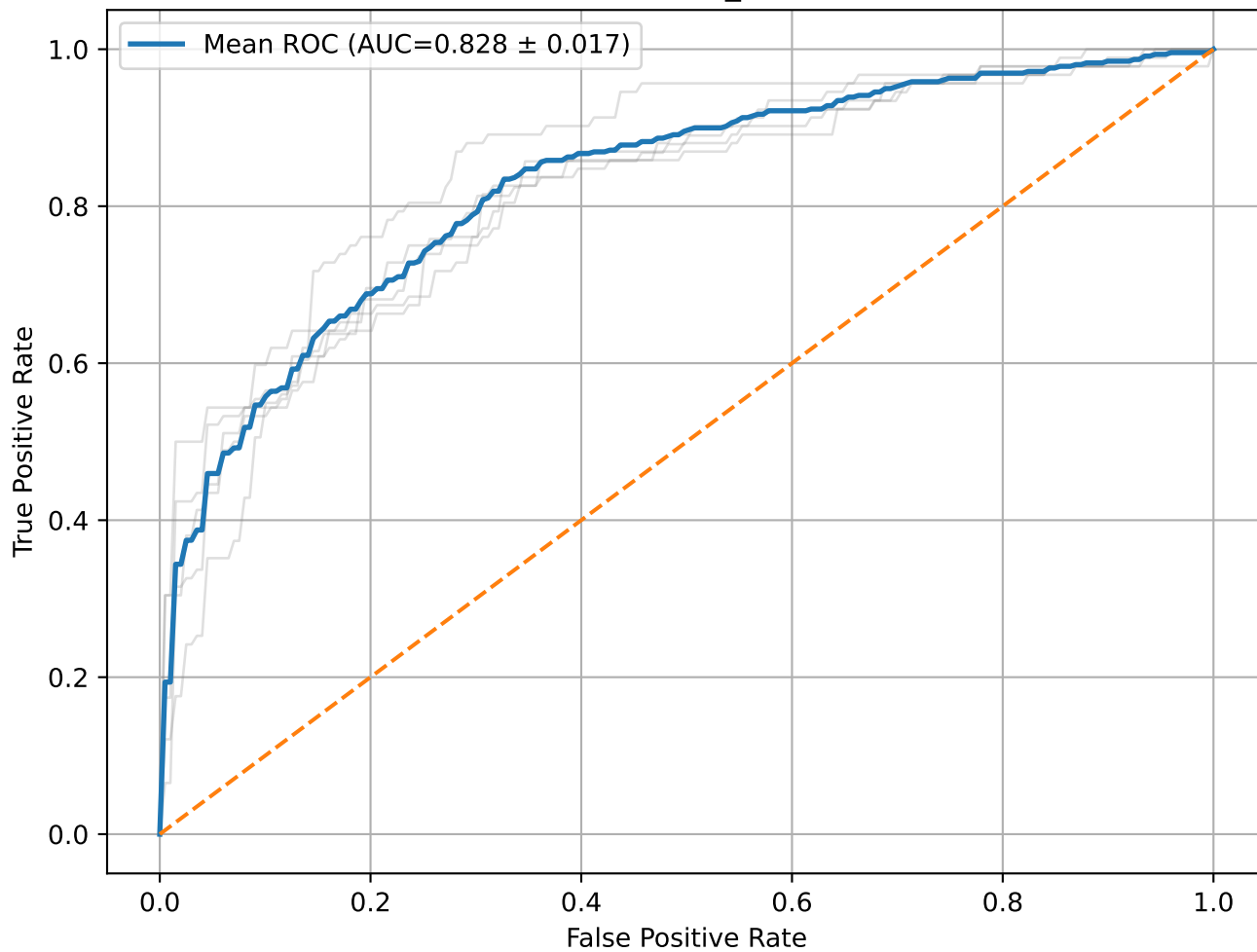
Confusion Matrix - BAGGING (last fold)



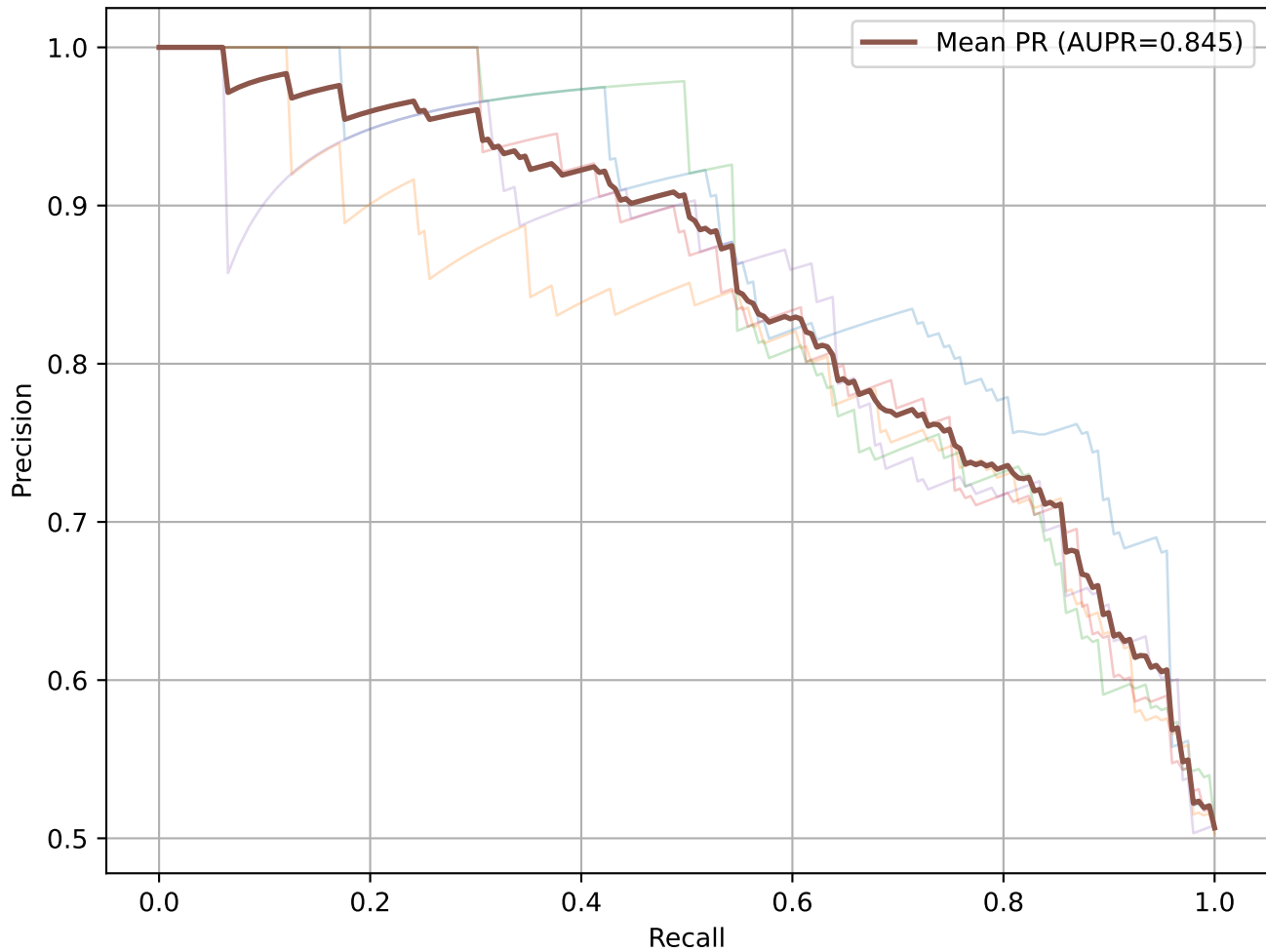
ClassificationReport - BAGGING (last fold)

	precision	recall	f1-score	support
0	0.84	0.85	0.84	89
1	0.86	0.84	0.85	92
accuracy			0.85	181
macro avg	0.85	0.85	0.85	181
weighted avg	0.85	0.85	0.85	181

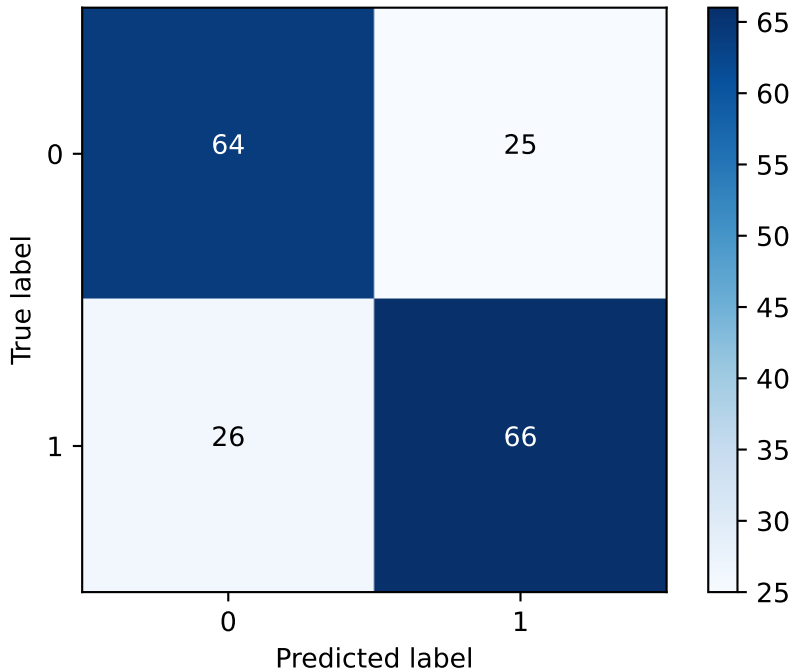
ROC - PCA_SVM



The figure displays a Receiver Operating Characteristic (ROC) curve for the proposed method. The x-axis represents the False Positive Rate (FPR) and the y-axis represents the True Positive Rate (TPR), both ranging from 0.0 to 1.0. The proposed method's performance is shown as a thick brown line, which is labeled in the legend as "Mean PR (AUPR=0.845)". This curve is significantly higher than the diagonal line, indicating good predictive performance. Several other methods are represented by thinner, colored lines (blue, green, orange, purple, pink), all of which fall below the proposed method's curve, indicating inferior performance. The legend in the top right corner identifies the brown line as the "Mean PR (AUPR=0.845)".



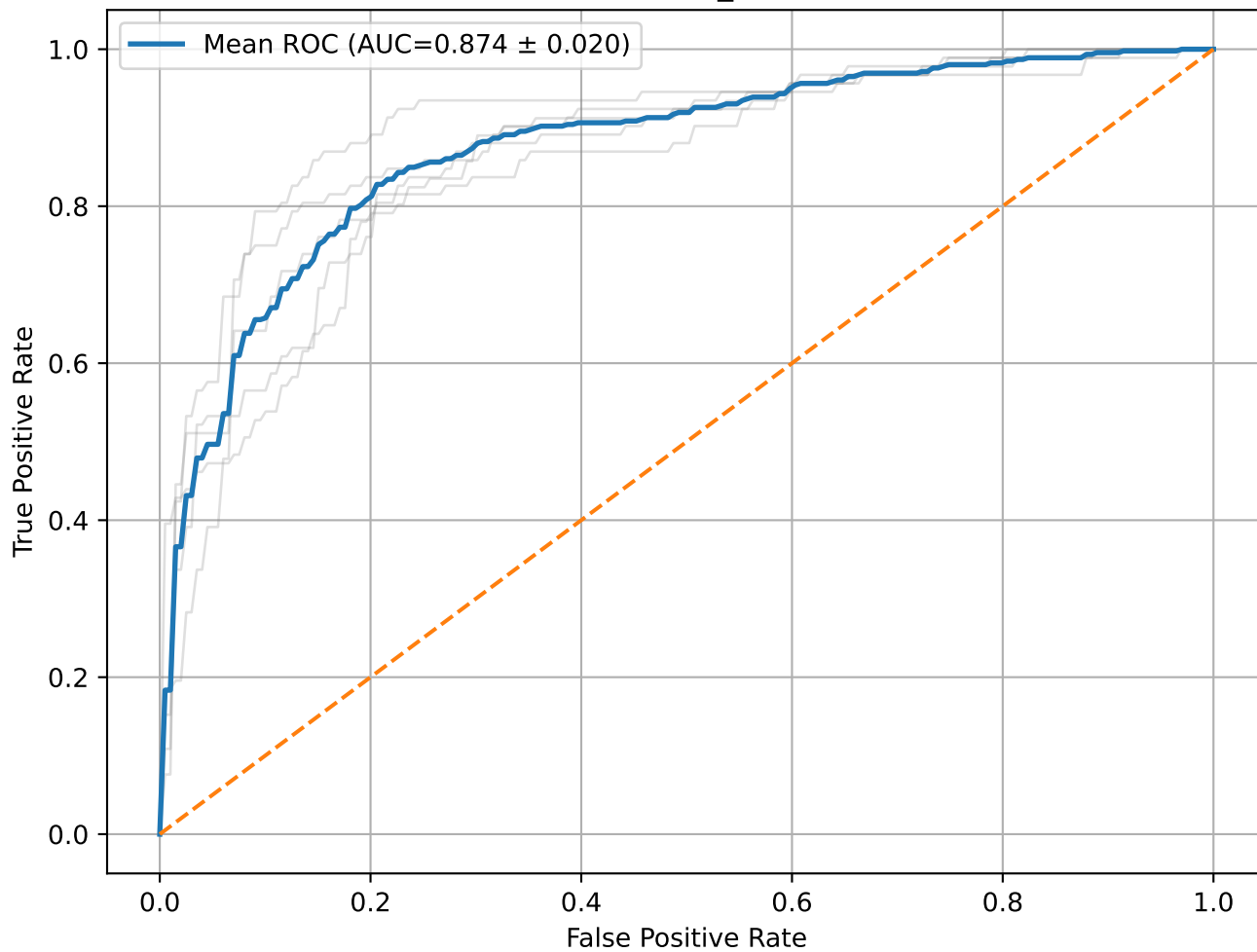
Confusion Matrix - PCA_SVM (last fold)



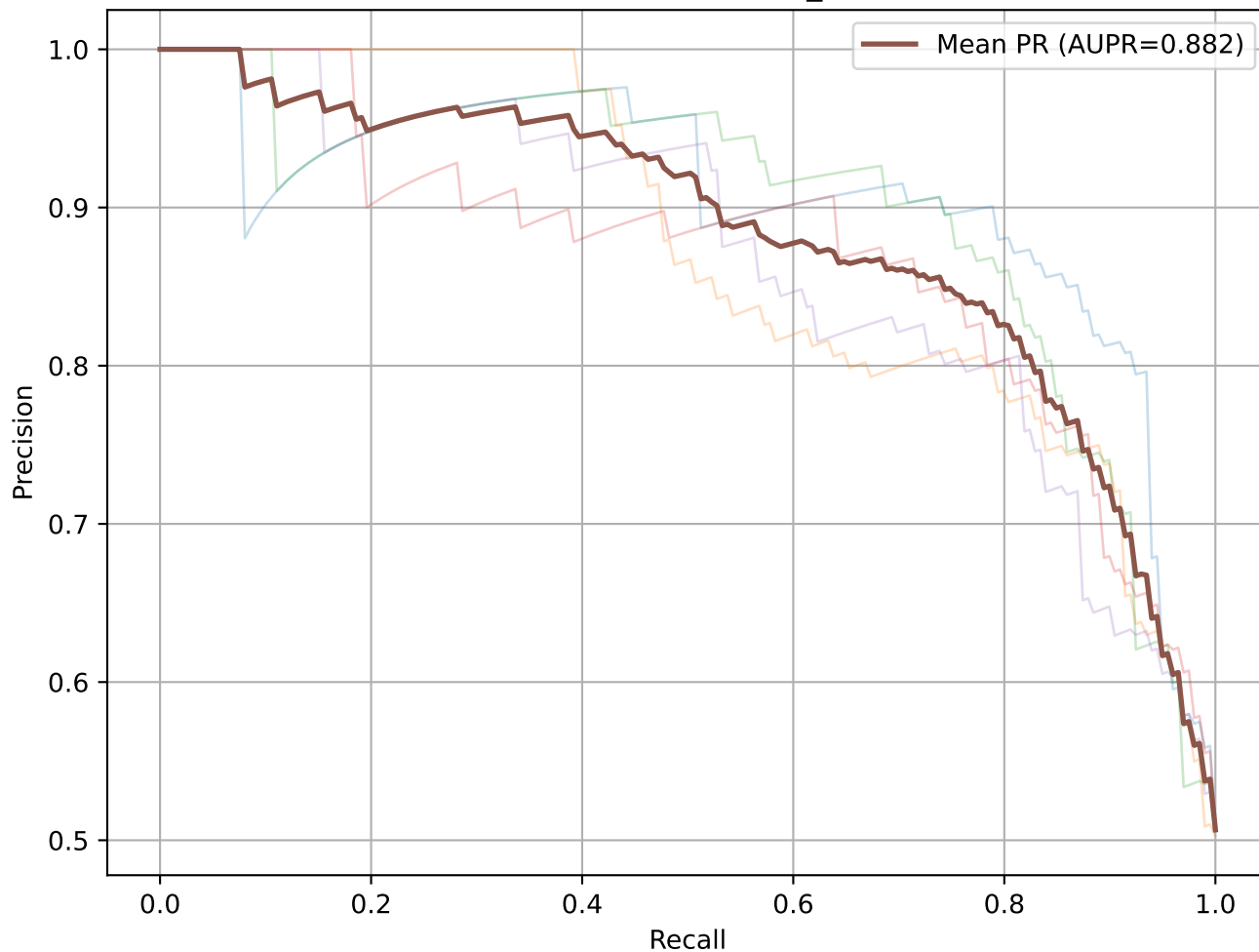
ClassificationReport - PCA_SVM (last fold)

	precision	recall	f1-score	support
0	0.71	0.72	0.72	89
1	0.73	0.72	0.72	92
accuracy			0.72	181
macro avg	0.72	0.72	0.72	181
weighted avg	0.72	0.72	0.72	181

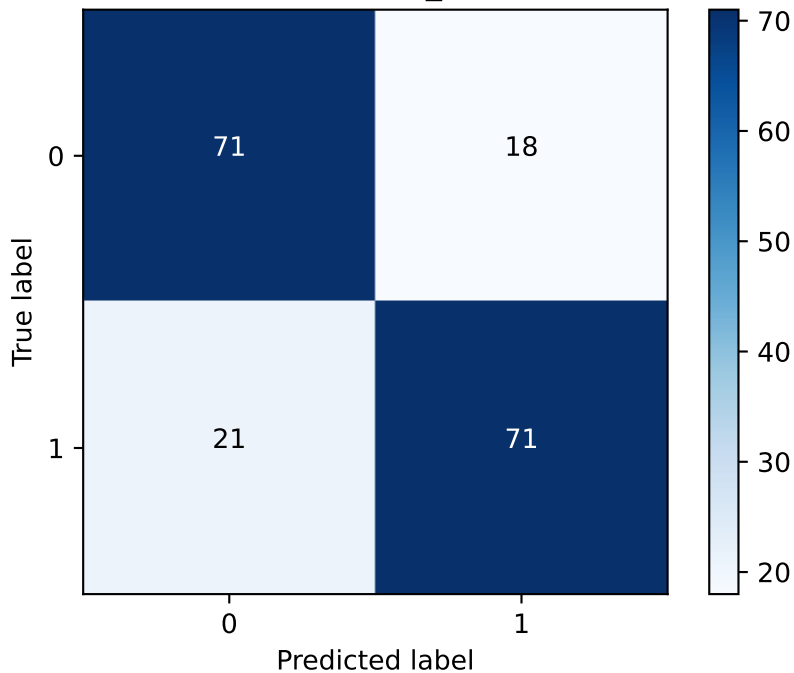
ROC - GA_XGB



Precision-Recall - GA_XGB



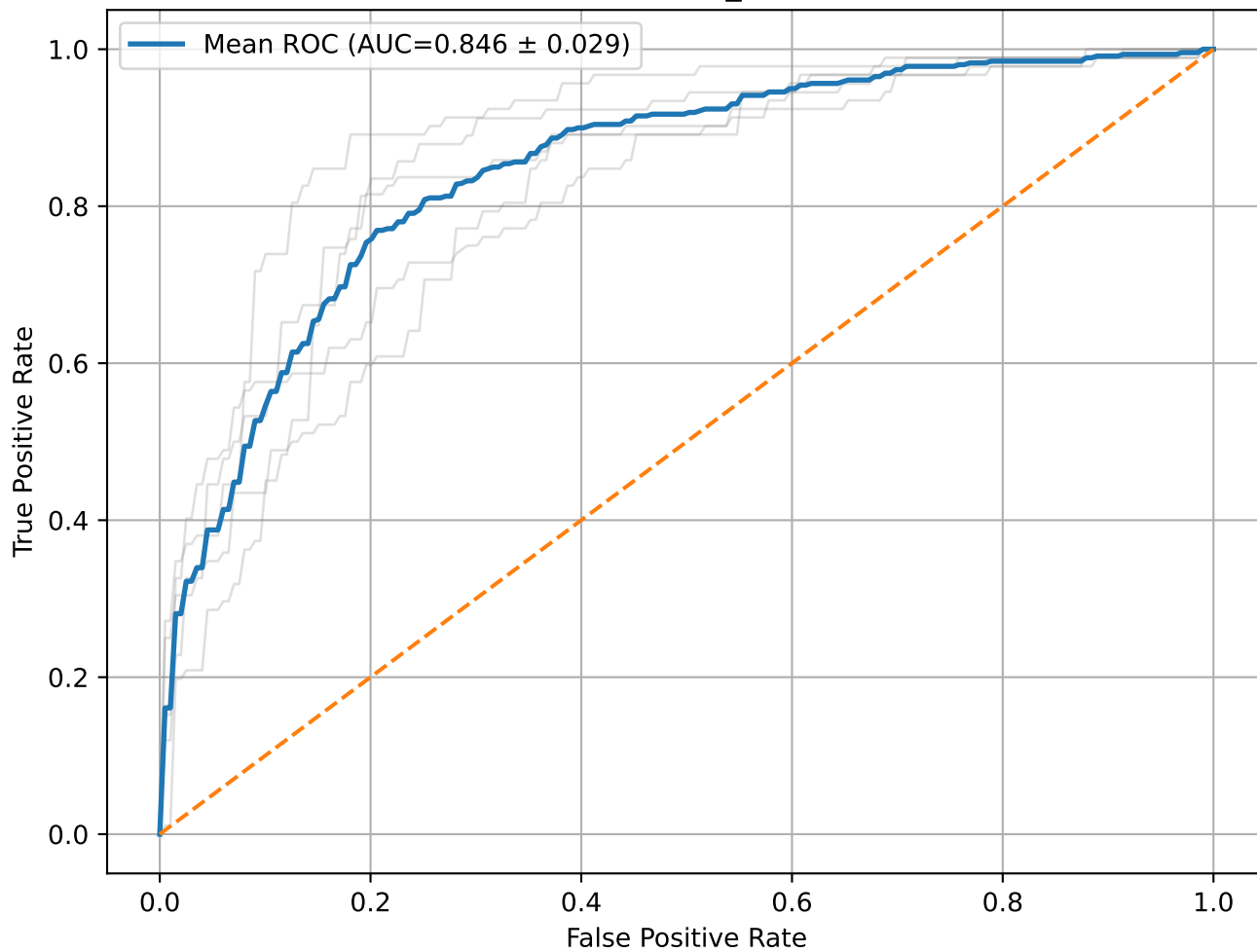
Confusion Matrix - GA_XGB (last fold)



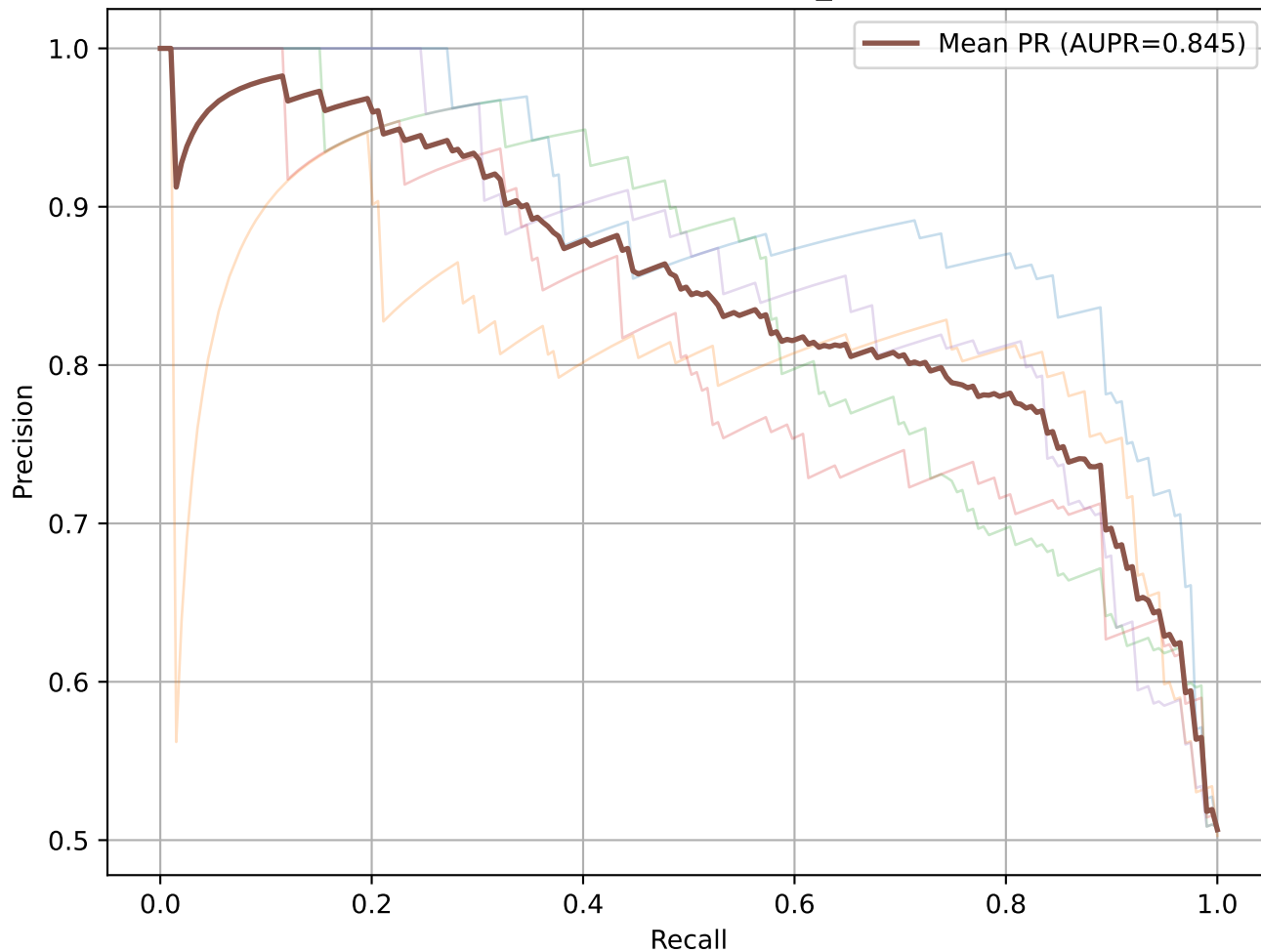
ClassificationReport - GA_XGB (last fold)

	precision	recall	f1-score	support
0	0.77	0.80	0.78	89
1	0.80	0.77	0.78	92
accuracy			0.78	181
macro avg	0.78	0.78	0.78	181
weighted avg	0.78	0.78	0.78	181

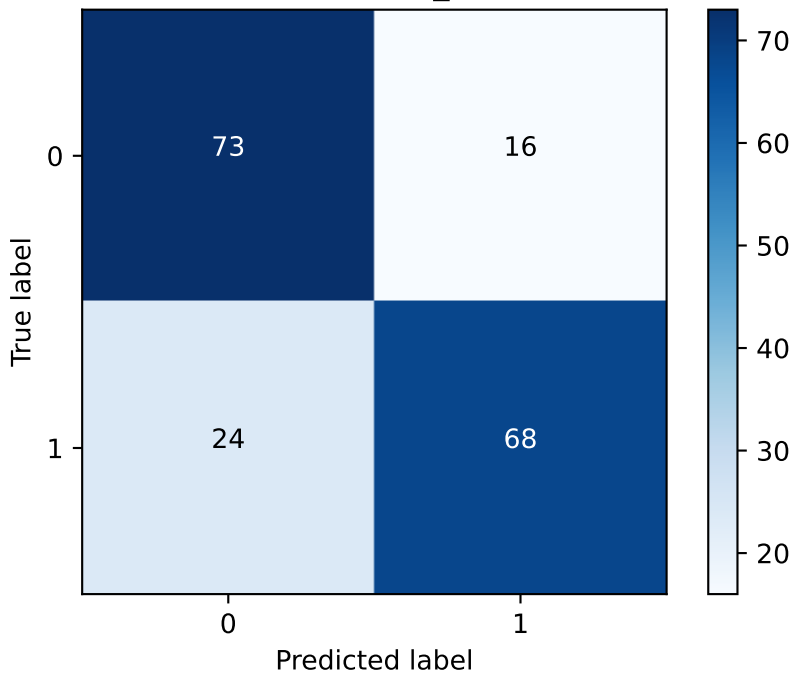
ROC - PSO_SVM



Precision-Recall - PSO_SVM



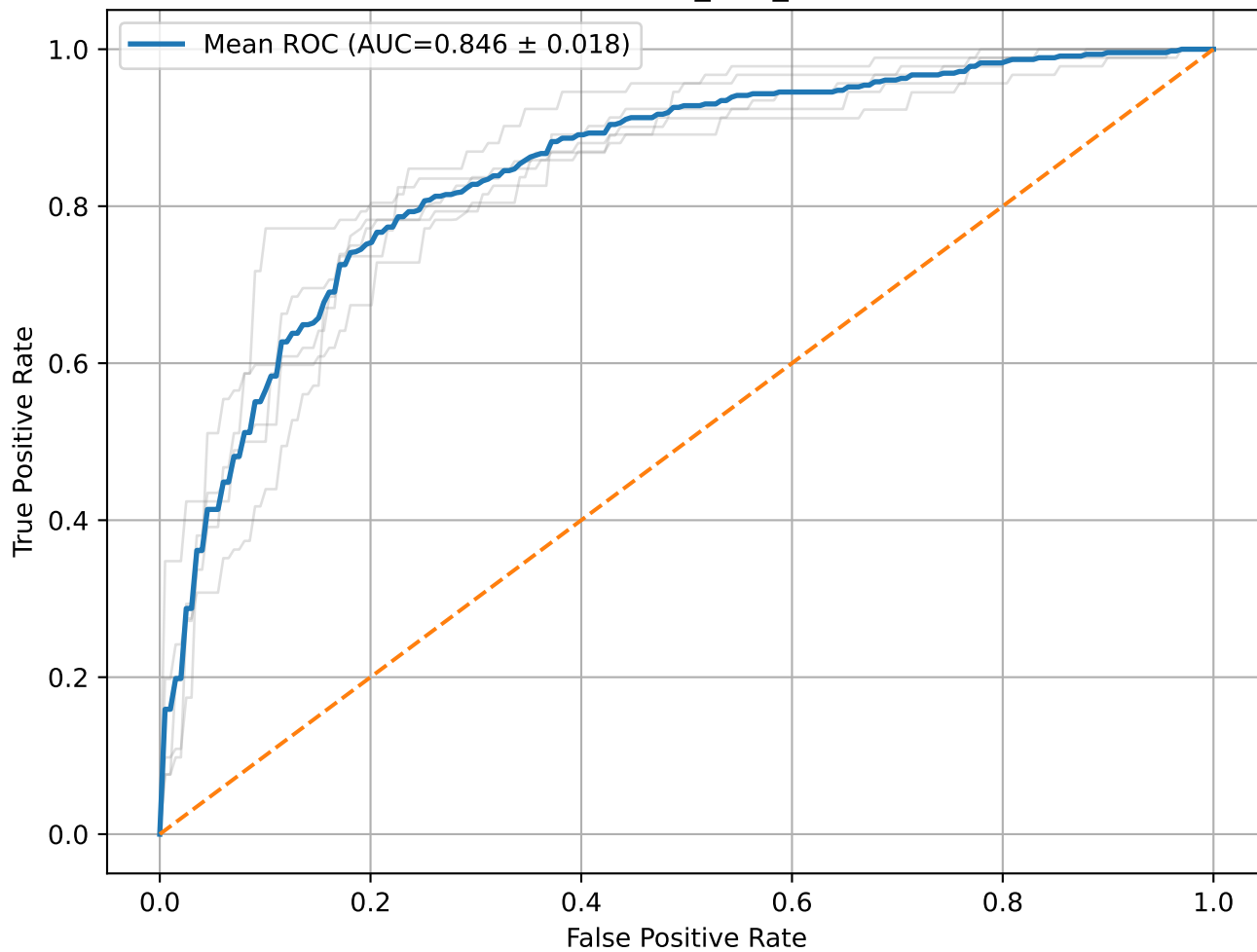
Confusion Matrix - PSO_SVM (last fold)



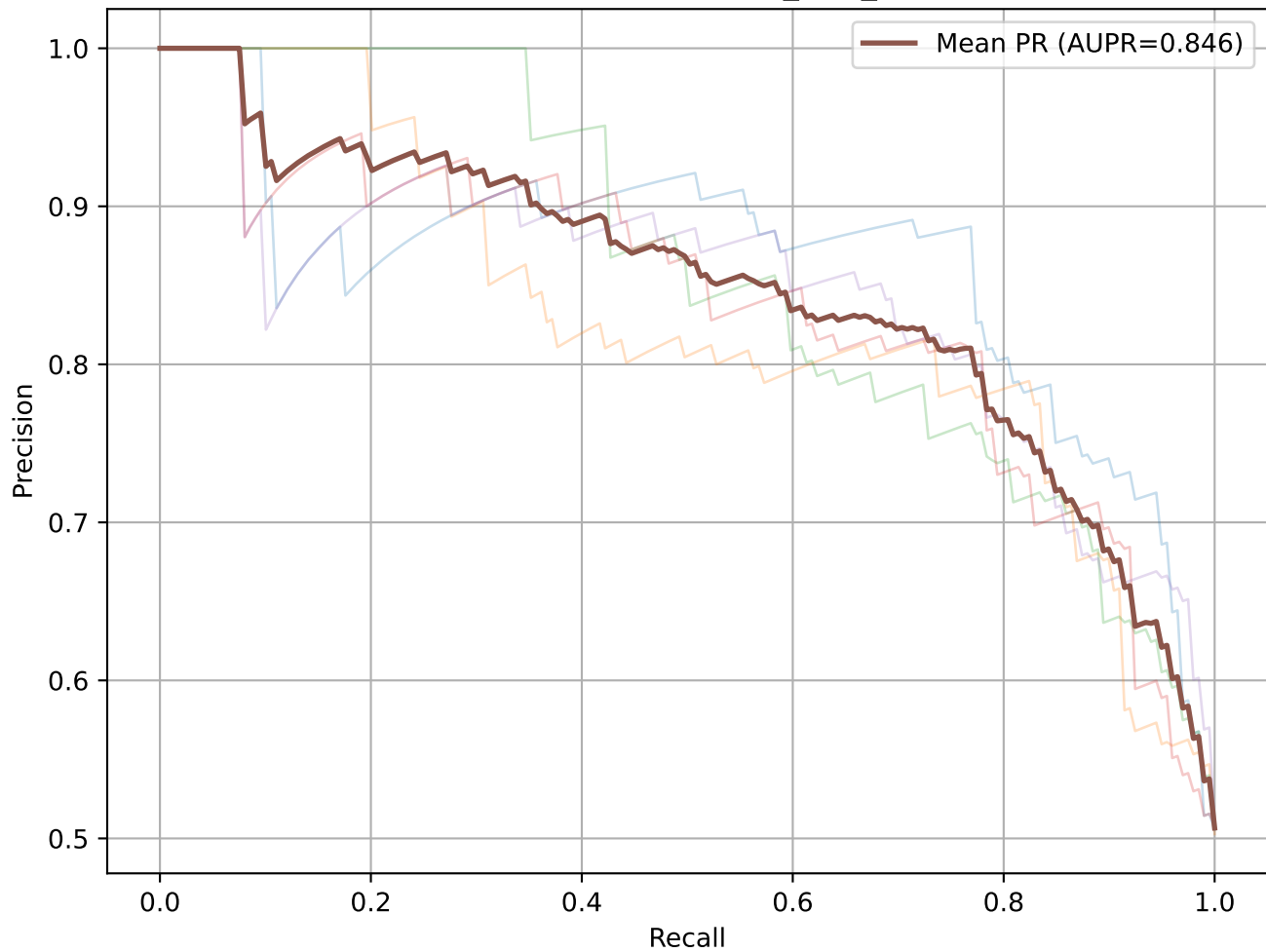
ClassificationReport - PSO_SVM (last fold)

	precision	recall	f1-score	support
0	0.75	0.82	0.78	89
1	0.81	0.74	0.77	92
accuracy			0.78	181
macro avg	0.78	0.78	0.78	181
weighted avg	0.78	0.78	0.78	181

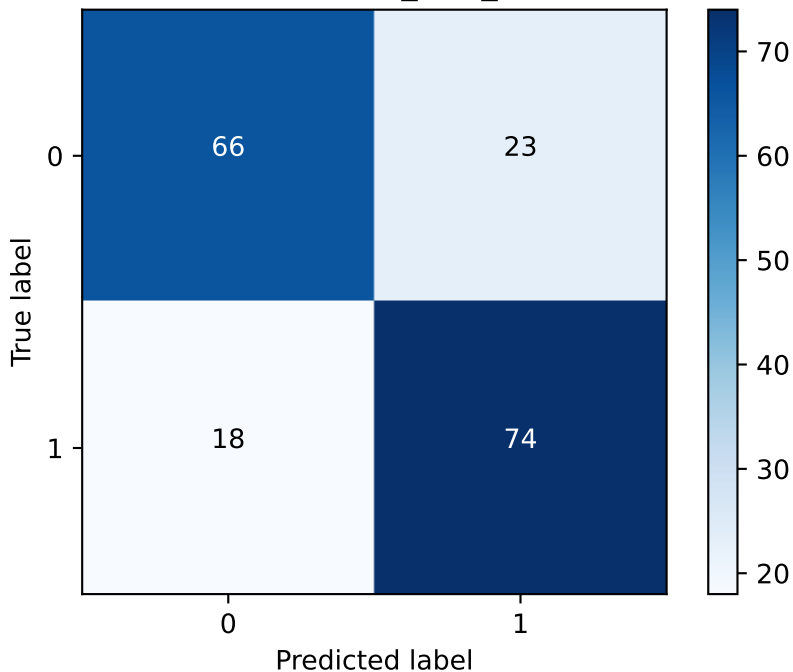
ROC - BAYES_OPT_SVMS



Precision-Recall - BAYES OPT SVMS



Confusion Matrix - BAYES_OPT_SVMS (last fold)



ClassificationReport - BAYES_OPT_SVMS (last fold)

	precision	recall	f1-score	support
0	0.79	0.74	0.76	89
1	0.76	0.80	0.78	92
accuracy			0.77	181
macro avg	0.77	0.77	0.77	181
weighted avg	0.77	0.77	0.77	181

Model Comparison Table (sorted by mean ROC AUC)

model	mean_roc_auc	std_roc_auc	mean_aupr	mean_accuracy	mean_f1
lgb	0.899381	0.017253	0.903563	0.843240	0.844769
stacking	0.886845	0.020077	0.897458	0.835481	0.835270
voting	0.885303	0.022165	0.896241	0.828869	0.828583
bagging	0.877502	0.021191	0.885519	0.814522	0.820344
ga_xgb	0.874716	0.019772	0.881672	0.811214	0.810759
xgb	0.872322	0.020279	0.878767	0.818936	0.820326
rf	0.858949	0.030408	0.877424	0.782515	0.781916
svm	0.852853	0.028208	0.849090	0.791336	0.796657
bayes_opt_svms	0.846777	0.018315	0.845752	0.778131	0.786061
pso_svm	0.846045	0.029470	0.844643	0.772558	0.774146
logistic	0.845200	0.020866	0.846017	0.781410	0.787811
pca_svm	0.828213	0.017413	0.844634	0.742790	0.742953
dnn	0.761015	0.034039	0.735772	0.690857	0.677713

Combined ROC - All Models

